Multigrid Self-Gravity in the AthenaK Shearing Box: I. Mesh-Refinement Infrastructure (Phase 0), Shear-Periodic Boundary Conditions (Phase 1), and Gravity on Refined Meshes (Phases 2a–2b)

`athenak-multigrid` tree, branch `multigrid`

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Abstract

This document opens the *multigrid track* of the self-gravity program: the route to gas + dust self-gravity in AthenaK shearing boxes *with mesh refinement.* It records Phase 0 — the shearing-box/refinement infrastructure every later phase rests on: (i) an `orbital_advection` toggle with a fully consistent non-FARGO (full-velocity) mode; (ii) an enforced *interior-x_1* refinement policy replacing AthenaK's blanket shearing-box+refinement prohibition; (iii) orbital advection (FARGO) on refined meshes under an *annular* refinement policy (refined regions span the full azimuthal extent); and (iv) *adaptive* refinement (Phase 0c): the shear-boundary communication state is rebuilt after every AMR update, a graded refinement cap keeps 2:1 balancing from ever refining the shear-periodic boundary columns, and an upstream derefinement-ordering bug that split annular rings is fixed. Validation: a refined-ring run reproduces a reference SMR run bit-for-bit in non-FARGO mode; with FARGO on the ring, mass is conserved exactly and a 20-orbit epicycle test tracks the analytic solution with an amplitude drift of only 0.008% (versus 0.064% on the uniform grid), with the epicyclic amplitude invariant flat to 10^{-4}. Under AMR, a prescribed moving ring that is created and destroyed every orbit for 20 orbits (2296 MeshBlocks created, 2240 destroyed) conserves mass exactly and matches the static-ring run an order of magnitude more closely than either matches the uniform grid; on the vortical-shwave problem the moving-ring solution converges to the static-ring one (4.7×10^{-2} apart at 64^2, 1.8×10^{-4} at 128^2). Phase 1 (§6) then makes the multigrid Poisson solver itself shear-aware: shear-periodicity is imposed purely as a boundary *identification* — a y-remapped x_1 ghost fill applied at every level of the multigrid hierarchy through global boundary planes, with the 7-point operator untouched. Validated against the FFT oracle: solves at shear-periodic instants are bit-identical to plain-periodic solves; the analytic shearing-wave potential is recovered with errors *below* the FFT solver's own (9.5×10^{-8} vs. 1.6×10^{-6} at 128^3); the swing-amplification history matches the FFT run to 2×10^{-7}; and the V-cycle contraction factor — the one open risk named in Phase 0 — degrades only from 0.116 to 0.157 per cycle at a near-worst fractional shift, reaching the same 10^{-11} roundoff floor. Phase 2a (§7) puts the solver on *refined* shearing-box meshes with interior-ring static refinement. The never-exercised octet machinery inherited with Athena++ #776 is first anchored on the isolated binary-potential test of Tomida & Stone §4.2 (region-wise second order against the analytic two-sphere solution); opening the shear machinery to multilevel meshes whose x_1 boundary blocks remain at the root level then requires *no new shear-fill code* — only a relaxed guard. Validation: sheared-instant solves on the refined mesh are bit-identical to the periodic twin; the analytic shearing-wave error converges at second order in L_2 (first order at the max, which sits on the refinement interface exactly as in the periodic case); the swing test on a level-1 ring matches its uniform twin to the hydro truncation level — the gravity-off control deviates *more* than the gravity-on run — and multi-rank solves reproduce

serial to a float32 ulp. Phase 2b (§8) then admits refinement *on* the shear boundary (boundary annuli): the octets tiling the shear faces receive a y-remapped x_1 ghost fill — the Phase-1 plane algorithm at octet granularity, in host code with no MPI seam (octets are rank-replicated) — through scalar per-face twins of the validated remap kernels, and the block shear planes are sized at the boundary blocks' level. Annuli solves at shear-periodic instants are bit-identical to their periodic twins at one and two refinement levels; the generic-phase L_2 equals the interior-ring value to four digits (the meshes differ by a half-box shift); contraction and floors match the periodic twins; and the annuli swing run matches its uniform twin to hydro truncation (again dominated by the gravity-off control) at machine-level phase lock. Phase 2c (§9) admits *adaptive* refinement: the shear planes, offsets, and octet flags rebuild after every remesh inside the existing `PrepareForAMR` path. A prescribed moving ring sweeping through the swing (392 MeshBlocks created, 280 destroyed) matches the uniform run to hydro regridding truncation with the gravity-off control dominating; the fixed-ring limit recovers machine-level phase lock and matches the static-SMR run to the float32 floor; and a 2-rank AMR run reproduces serial to every printed digit. The companion document `fft_selfgravity_validation.pdf` covers the FFT solver used as the machine-precision oracle throughout this program; a Julia notebook (`multigrid_validation_tests.ipynb`) reproduces every test here.

Contents

1 Plan and scope

1.1 Goal and roadmap

The scientific target is shearing-box simulations combining gas self-gravity, dust with mutual drag (streaming instability, including the background radial pressure gradient), and mesh refinement that zooms onto high dust-density regions in spiral arms launched by gravitational instability (cf. Baehr, Zhu & Yang 2022, ApJ 933, 100, who omitted the pressure gradient). Mesh refinement is the reason AthenaK was chosen, so the Poisson solver must ultimately live on refined meshes — which an FFT cannot. The adopted route therefore makes the *multigrid* solver shear-aware, with the validated FFT solver retained in three roles: machine-precision oracle for the multigrid shear boundary conditions, production solver for uniform-grid runs, and donor of exact open-z Dirichlet boundary values for stratified multigrid runs.

The phases of the multigrid track:

1. **Phase 0 (this document, done):** shearing box + refinement infrastructure. 0a: an `orbital_advection` on/off toggle with a consistent full-velocity (non-FARGO) mode, and

the interior-x_1 refinement policy. 0b: FARGO on refined meshes under the annular refinement policy. 0c: adaptive refinement — dynamic creation and destruction of refined rings, without gravity, isolating the mesh machinery from the solver before Phase 2 couples them. Without 0b, refined shearing boxes pay a timestep penalty of $\sim (c_s + q\Omega L_x/2)/c_s \approx 10$–30 for gravito-turbulence boxes.

2. **Phase 1 (this document, §6, done):** shear-periodic boundary conditions in the multigrid driver (`MultigridDriver`), validated against the FFT oracle: exactness at shear-periodic instants, shearing-wave accuracy at generic shear phase, and the swing-amplification test with `solver=multigrid`. The V-cycle convergence rate with remapped boundaries — the one open risk of the program — is measured: the per-cycle contraction factor degrades only from 0.116 to 0.157.

3. **Phase 2:** multigrid gravity on refined shearing-box meshes (SMR first, then AMR), compared against high-resolution uniform FFT references.

4. **Phase 3–5:** merge with the `athenak-dust` particle-mesh dust module (dust density in the Poisson source, gravitational force on particles, pressure-gradient source term for the streaming instability), stratification with open-z boundaries, and the science runs.

1.2 Phase-0 scope and restrictions

What works after Phase 0, all enforced by fatal errors rather than conventions:

- Hydrodynamic shearing boxes with SMR, under the *uniform-boundary-level* rule: both shear-periodic x_1 boundary faces must share one refinement level. Refinement interior in x_1 always satisfies this (faces stay at root); alternatively *both* boundary faces may be refined uniformly (full x_2/x_3 extent, same level) — physically natural for a structure at the boundary, whose shear-periodic images live on both faces.

- Orbital advection on refined meshes, provided every refined region spans the full x_2 (azimuthal) extent — *annular rings* (interior, or boundary annuli as above). Without FARGO (`orbital_advection=false`) arbitrary refinement shapes are allowed subject only to the boundary-level rule.

- Adaptive mesh refinement (Phase 0c). The shear-boundary communication state (`ShearingBox::x1bndry_mbg` GID lists, MPI request arrays, CC/FC buffers) is rebuilt after every AMR update; AMR flag handling keeps rings intact (refine a ring if any member is flagged, derefine only if all are) and a *graded cap* refuses any refinement whose 2:1 balancing would propagate into the shear-periodic boundary columns (§5.8).

Not yet supported/exercised: MHD orbital advection with refinement (the face-centered EMF remap at level interfaces needs a $\nabla \cdot B$ analysis; fatal-guarded), multi-rank (MPI) exercise of the AMR rebuild path (the code is rank-aware but all Phase 0c runs are serial), and a production-size demonstration of the FARGO speedup itself. SMR/AMR runs must write `bin` outputs — the vtk writer does not support refined meshes.

2 Numerical method

2.1 Two frames for the shearing box

With background shear flow $\mathbf{v}_0 = -q\Omega x\,\hat{y}$, AthenaK can now evolve either of two variable sets, selected by <shearing_box> orbital_advection:

FARGO frame (`orbital_advection = true`, default). The stored velocity is the *fluctuation* $\mathbf{v}' = \mathbf{v} - \mathbf{v}_0$; the background advection is applied analytically once per step by the orbital-advection shift (Sec. 2.2). The momentum source terms are

$$\dot{M}_x = 2\Omega\,\rho v'_y, \qquad \dot{M}_y = -(2-q)\,\Omega\,\rho v'_x, \tag{1}$$

where the $(2-q)$ coefficient is the sum of Coriolis (-2Ω) and the advection of background shear by the radial fluctuation $(+q\Omega)$. The CFL condition sees only $|\mathbf{v}'|$.

Full-velocity frame (`orbital_advection = false`). The stored velocity includes the shear. The source terms are the full Coriolis force plus the tidal force, and (for an ideal EOS) the tidal work:

$$\dot{M}_x = 2\Omega\,\rho v_y + 2q\Omega^2 x\,\rho, \qquad \dot{M}_y = -2\Omega\,\rho v_x, \qquad \dot{E} = 2q\Omega^2 x\,\rho v_x \tag{2}$$

(the Coriolis force does no work). Because the wrapped data at the shear-periodic x_1 faces comes from the opposite side of the box, where the background azimuthal velocity differs by $q\Omega L_x$, the boundary exchange must offset the wrapped azimuthal momentum and energy:

$$M_y \to M_y + \rho\,\Delta v, \qquad E \to E + M_y\,\Delta v + \tfrac{1}{2}\rho\,\Delta v^2, \qquad \Delta v = \pm\,q\Omega L_x \tag{3}$$

(+ at the inner face, − at the outer). Initial conditions must include $\mathbf{v}_0$. The two frames are the same physics on different variables; comparing them is a strong consistency test (Sec. 5). The cost of the full-velocity frame is the timestep,

$$\frac{\Delta t_{\rm FARGO}}{\Delta t_{\rm full}} \approx \frac{c_s + q\Omega L_x/2}{c_s + |\mathbf{v}'|_{\max}}, \tag{4}$$

measured directly as 5.7× for the $L_x = 2\pi$ swing-test box (Sec. 5.1) and reaching 10–30× for gravito-turbulence boxes — plus the extra truncation diffusion of advecting structures through the grid at many c_s. Non-FARGO is therefore the correctness baseline, not the production mode.

2.2 Orbital advection and its refinement problem

The orbital-advection step shifts every x-column of every MeshBlock by the background displacement $\delta y(x) = q\Omega\,x\,\Delta t$, as an integer number of cells plus a conservative fractional remap,

$$q_j \;\to\; q_{\,j-j_{\rm off}} - \big(F_{j+1-j_{\rm off}} - F_{j-j_{\rm off}}\big), \qquad j_{\rm off} = \left\lfloor \frac{\delta y}{\Delta y} \right\rfloor, \tag{5}$$

with upwind remap fluxes F of first (`dc`), second (`plm`), or third (`ppmx`) order — the same kernels as the shear-periodic boundary remap. The required y-ghost data travels in buffers of depth n_g + `maxjshift` exchanged with the two x_2-face neighbors.

The crucial observation for refinement: *the shift kernel is already per-block correct on a multilevel mesh.* Each block computes δy from its own cell centers and applies it in its own Δy; blocks at different levels shift by the same physical displacement expressed in their own cell units, and after the shift the ordinary AMR ghost exchange (with prolongation/restriction at x_1/x_3 level interfaces) restores consistency to interpolation order, since a uniform-in-y translation commutes with the interpolations to that order. What is *not* level-safe is the communication pattern: the buffer exchange assumes both x_2-face neighbors are same-level, same-size blocks, which fails at any coarse–fine interface in y. This is exactly why refinement previously worked only with orbital advection disabled.

2.3 Refinement policies

Rather than generalizing the y-communication to level interfaces, Phase 0 removes the interfaces:

Uniform-boundary-level policy (always). All MeshBlocks on the two shear-periodic x_1 boundary faces must share *one* refinement level. The reason is structural: the unshifted wrap across x_1 goes through the ordinary neighbor machinery (the tree treats `shear_periodic` exactly like `periodic`), and the shear pass then redistributes the wrapped ghost data azimuthally *within each face* — with per-block Δy arithmetic and a level-aware target lookup (`FindTargetMB` shifts lx_2 modulo the block count at the source block's own level). Nothing in that machinery references the root grid; its only real assumption is same-size blocks along each face and a same-level wrap between the faces. Hence the common level may exceed root: refining *both* boundary faces uniformly (full x_2/x_3, same level) is legal, and a mixed-level boundary is a fatal error. Enforced block-by-block at tree build and restart (`Mesh::CheckShearingBoxRefinement`). Truly mixed-level shear boundaries (a refined patch on one face wrapping onto coarse blocks of the other) would require combining the y-shift with prolongation/restriction in the shear comms — deferred; no production code implements it.

Annular policy (when FARGO is on). Every refined region must span the full x_2 extent over its x_1–x_3 footprint: refinement comes in *rings*. Within a ring every x_2-face neighbor is same-level and same-size, so the per-level shift plus the *existing* communication code are valid with no changes. Enforced by ring-completeness counting over (level, lx_1, lx_3); a partial ring with FARGO on is a fatal error naming the offending ring. For AMR, refine/derefine flags are synchronized ring-wise after the global flag exchange — refine the whole ring if any member is flagged, derefine only if every member is — deterministically on all ranks. Scientifically the ring shape is a good match for the target problem: GI spiral arms are azimuthally extended, and a ring at the arm's radius captures the whole arm and every clump in it.

Buffer-depth safety. The integer shift on a refined block at radius x is $q\Omega|x|\Delta t/\Delta y_\ell$ cells, which the construction-time estimate `maxjshift` $= \lfloor \text{cfl} \cdot q\Omega \max|x| \rfloor + 1$ (now including the previously missing $q\Omega$ factor, doubled on multilevel meshes) may underestimate in unusual configurations. The unpack step therefore re-verifies the actual worst-case shift against the communicated buffer depth every step and aborts with a diagnostic rather than silently corrupting data.

3 What was built

Changes live in commits `e7ef37f2` (infrastructure) and `f1468da3` (bin outputs + docs); Phase 0a is ported from the `athenak` branch `test/shwave-smr`, with provenance kept in the input-file comments. Phase 0c adds commits `f0acd383` (derefinement-sort fix), `b5388c02` (AMR rebuild machinery, graded cap, moving-ring test), and `097a97ee` (vortical-shwave AMR test).

File (under `src/`)	Change
`hydro/hydro.cpp`, `mhd/mhd.cpp`	`orbital_advection` toggle; MHD+refinement fatal
`shearing_box/shearing_box.{hpp,cpp}`	store the toggle; `SetX1BndryMBs` / `ReinitAfterMeshUpdate`: rebuild boundary GID lists, MPI requests, and (via virtual `AllocateBuffers`, grow-only) CC/FC buffers after every AMR update
`shearing_box/shearing_box_srcterms.cpp`	dual-frame source terms, Eqs. (1)–(2)
`shearing_box/shearing_box_cc.cpp`	non-FARGO BC momentum/energy offset, Eq. (3)
`shearing_box/orbital_advection.cpp`	`maxjshift` estimate: $q\Omega$ factor, $2\times$ multilevel headroom
`shearing_box/orbital_advection_cc.cpp`	runtime shift-vs-buffer-depth guard (fatal)
`mesh/mesh.cpp`	blanket prohibition $\rightarrow$ policy warning
`mesh/build_tree.cpp`	`Mesh::CheckShearingBoxRefinement`: interior-x_1 and ring completeness
`mesh/mesh_refinement.{hpp,cpp}`	boundary-block refinement veto + graded cap, Eq. (6); ring-wise AMR flag sync; `ReinitAfterMeshUpdate` calls after each update; derefinement-sort end-iterator fix (upstream bug)
`pgen/tests/shwave.cpp`, `swing.cpp`	background shear in ICs when `orbital_advection=false`; `MovingRingRefine` user AMR criterion (`amr_moving_ring`)
`inputs/shearing_box/` (repo root)	`*_smr*.athinput`, `epicycle*.athinput`, `*_amr_ring*.athinput` test inputs (`bin` outputs)

Runtime options (`<shearing_box>` block): `orbital_advection = true|false` (default `true`). Everything else is policy, enforced automatically. The gravity solver selection of the companion document (`<gravity> solver = multigrid|fft`) is unaffected by any of this; Phase 1 will add `shear_periodic` boundary support to the multigrid driver behind the same input block.

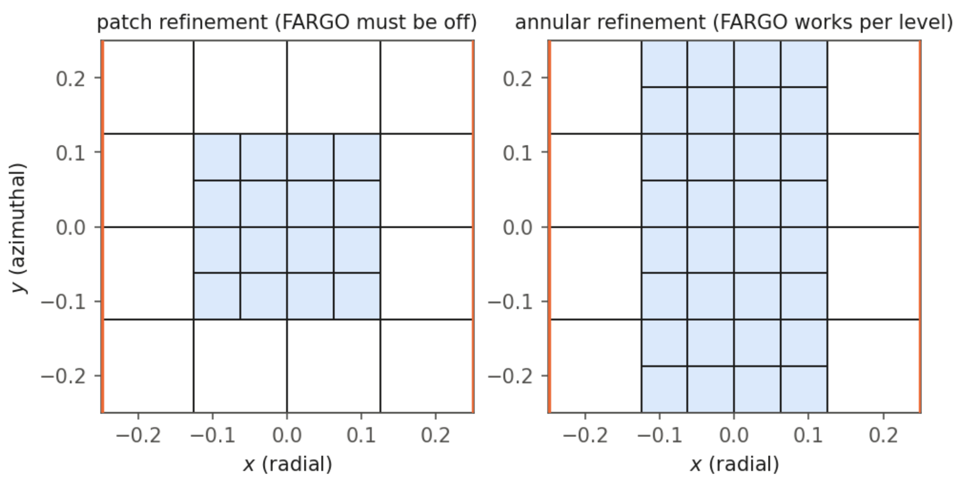

Figure 1: The two refinement layouts, drawn from the `bin` output block structure of actual runs (orange lines: shear-periodic x_1 boundaries; shaded: level-1 blocks). Left: a patch interior in both x_1 and x_2 — legal only with `orbital_advection=false`, because the coarse–fine interfaces in y break the shift communication. Right: the annular layout — the ring has no level jump in y, so FARGO runs per level with unmodified communication.

4 Building and compiling

Phase 0 needs no external dependencies beyond AthenaK's vendored Kokkos:

```
cd athenak-multigrid
git submodule update --init kokkos     # once
mkdir build && cd build
cmake ..                               # add -D Athena_ENABLE_FFT=ON for the FFT
make -j8                               #   oracle / swing+gravity cross-checks
```

Notes: SMR/AMR runs need `<mesh> nghost` ≥ 4 (even, for prolongation); the FFT flag is only required for the tests that put gravity in the loop (it fetches kokkos-fft and needs `brew install fftw` on macOS, as detailed in the companion document). All test inputs referenced below are under `inputs/shearing_box/` and `inputs/tests/`; runs in this document were executed from `validation/run/` and `validation/run_epicycle/`.

5 Tests and results

5.1 Frame consistency: FARGO vs. full-velocity, uniform grid

```
ATH=build/src/athena
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_nofargo \
     shearing_box/orbital_advection=false
# with gravity in the loop (requires FFT build):
$ATH -i inputs/tests/swing_selfgrav.athinput job/basename=SwingNoFargo \
     shearing_box/orbital_advection=false
```

The two frames agree on the frame-independent diagnostics of the JG05 vortical shearing wave — mass exactly, x-kinetic-energy history to 3.3% (the truncation gap between two different discretizations of the same problem). With self-gravity, the non-FARGO swing test tracks the linearized shearing-sheet solution with $L_2(\delta) = 1.2 \times 10^{-2}$ (FARGO: 0.9×10^{-2}) and phase-locking of the density to the instantaneous shearing wavevector at the 10^{-12} level; it costs 5.7× more cycles, exactly Eq. (4).

5.2 Refinement without FARGO (Phase 0a)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     shearing_box/orbital_advection=false
```

An interior level-1 patch (Fig. 1, left) in the JG05 vortical-shwave problem. The run reproduces the reference run from the original `test/shwave-smr` branch **bit-for-bit** in every history column — the port changed nothing.

5.3 Guard checks (must abort)

```
# refined region snapping to a shear-boundary block: interior-x1 policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     refined_region1/x1min=-0.24 refined_region1/x1max=-0.13
# partial-x2 ring with FARGO on: annular policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
```

```
     shearing_box/orbital_advection=true
```

Both abort with explanatory fatal errors; the second names the offending ring ("ring at level=3 lx1=2 lx3=0 has 4 of 8 MeshBlocks").

5.4 FARGO on an annular ring (Phase 0b)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
     job/basename=ring_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
     job/basename=ring_nofargo shearing_box/orbital_advection=false
```

The full-x_2 level-1 ring of Fig. 1 (right). With FARGO active on the refined mesh: mass conserved exactly (drift 0.0); x-kinetic-energy history agrees with the non-FARGO twin to 3.8% — the same truncation-level gap the two frames show on uniform grids — and with the uniform-grid FARGO run to 4.1%.

5.5 Refined x_1 boundary faces

The uniform-boundary-level rule allows refinement *at* the shear-periodic boundaries when both faces are refined uniformly. The test input refines the leftmost and rightmost root columns to level 1 over the full x_2/x_3 extent (boundary annuli), interior at root, orbital advection on:

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
     job/basename=epi_bndry problem/ipert=1
# guard (must abort): refine only ONE face -> mixed levels across the wrap
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
     refined_region2/x1min=0.0 refined_region2/x1max=0.12
```

Results: the epicycle matches the analytic solution to 5.0×10^{-6} — *identical to the last digit* with the uniform-grid and interior-ring values, so the level-1 shear-periodic wrap and the level-1 azimuthal redistribution preserve uniform fields exactly. The 20-orbit endurance run on this mesh (`epicycle_smr_bndry.athinput`) is *bit-identical* to the interior-ring endurance run over all 202 history outputs: for a uniform field the refined shear-boundary wrap contributes exactly nothing, the strongest available statement for this configuration. The vortical shwave conserves mass exactly and agrees with the uniform run to 4.9% in x-KE (interior ring: 4.1%) and with the interior-ring run to 0.9%; a three-panel density movie of uniform | interior ring | boundary annuli (`make_bin_movie_compare.jl`) shows the sheared wave crossing the refined boundary with no visible artifacts. The single-face guard aborts with a message naming the level span.

5.6 Two-level refinement (levels 1 and 2)

Both policies recurse in level, so nothing new is required for deeper hierarchies — verified with two configurations at levels 1+2 (8 root + 32 level-1 + 256 level-2 blocks each, FARGO on): nested interior rings (`hydro_incompress_shwave_smr_ring2.athinput`) and level-2 boundary annuli with level-1 buffer rings inboard (`hydro_incompress_shwave_smr_bndry2.athinput`). The epicycle matches the analytic solution to 5.0×10^{-6} on both meshes and the two runs are *bit-identical* to each other; the vortical shwave conserves mass exactly and agrees with the uniform run at truncation level (4.5% and 5.2% in x-KE; 0.45% between the two-level and one-level rings). The three-panel density movie (uniform | nested rings | two-level boundary annuli) shows no artifacts at any level interface or at the refined shear boundary.

5.7 The endurance test: epicycles for 20 orbits

The `ipert=1` epicycle has spatially uniform $v_x(t)$, $v'_y(t)$: an exact nonlinear solution that any correct y-shift must preserve *exactly*, making it the sharpest probe of the per-level shift and its ring-boundary communication — any indexing or buffer defect appears as $O(1)$ corruption. Two long runs ($L = 20$, $128^2 \times 4$ effective resolution, ideal EOS with $c_s = \sqrt{5/3}$, $v_x(0) = 0.1 = 0.077\, c_s$, 20 orbits = 20 epicyclic periods since $\kappa = \Omega$ at $q = 3/2$):

```
$ATH -i inputs/shearing_box/epicycle.athinput            # uniform 128x128x4
$ATH -i inputs/shearing_box/epicycle_smr_ring.athinput   # level-1 ring + FARGO
```

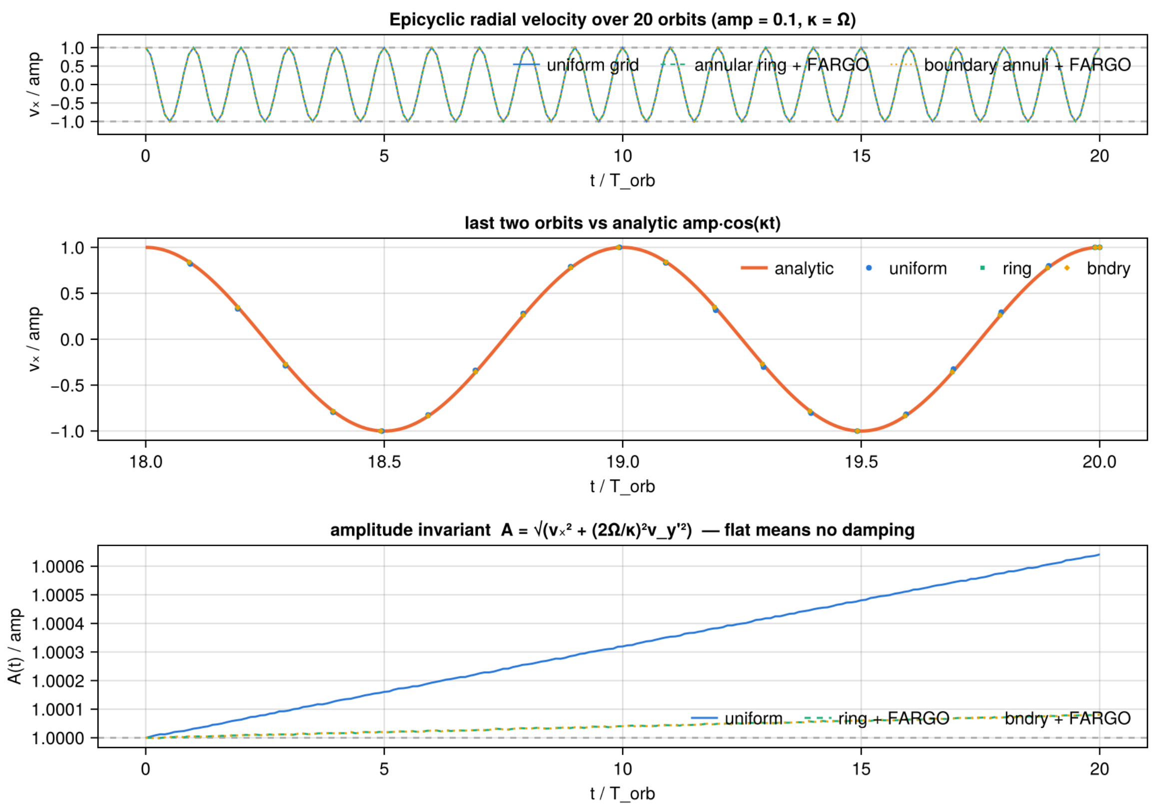

Figure 2: Volume-averaged radial velocity of the epicycle test over 20 orbits, normalized to the initial amplitude. Top: both runs oscillate between ± 1 with no visible decay. Middle: the final two orbits against the analytic $v_x = A \cos \kappa t$. Bottom: the epicyclic amplitude invariant $A = [v_x^2 + (2\Omega/\kappa)^2 v_y'^2]^{1/2}$, which isolates numerical damping/growth from phase error.

run	v_x range / amp	amplitude drift, 20 orbits	max $\lvert v_x - \text{analytic}\rvert$ / amp
uniform 128^2	$[-1.0006, +1.0005]$	$+0.064\%$	2.4%
annular ring + FARGO	$[-1.0000, +1.0001]$	$+0.008\%$	0.63%
boundary annuli + FARGO	*bit-identical to the annular ring, all 202 outputs*		

Both runs hold the ± 1 envelope; the amplitude invariant is flat to a few $\times 10^{-4}$ (uniform) and 10^{-4} (ring) — a slow secular *growth* characteristic of RK2 on an oscillator, not shearing-box or refinement physics. The 3.3% peak difference between the runs is pure phase drift: the ring run takes smaller steps (finer cells) and accumulates *less* RK2 phase error, which is why the refined-mesh FARGO run is the one closer to the analytic solution. A defect in the ring communication would instead show as amplitude noise — none is present at the 10^{-5} level.

5.8 AMR in the shearing box (Phase 0c)

AMR renumbers MeshBlock GIDs and re-assigns blocks to ranks on every mesh update, so the shear-boundary communication state built at construction — the `x1bndry_mbgid` GID lists, the MPI request arrays sized to them, and the send/recv buffers — goes stale. `ShearingBox::ReinitAfterMeshUpdate` now rebuilds all of it from the current MeshBlockPack, called from `AdaptiveMeshRefinement` after the post-update policy check and before any boundary exchange on the new mesh; the startup fatal guard is gone. Orbital advection needs no rebuild (its buffers are neighbor-indexed and sized to `max_nmb_per_rank` from the start).

Two defects surfaced by the multi-level tests, both now fixed, are worth recording. First, vetoing refinement of the boundary blocks themselves is not enough: a criterion refining blocks *near* the boundary can force refinement *of* the boundary columns through 2:1 balancing. The veto is therefore extended to a *graded cap*: a block at level $s \geq 1$ above the (uniform) boundary level may be refined only if there is room for the intermediate buffer columns between it and the boundary, which in integer logical coordinates at the block's own level reads

$$2^{s+1} - 1 \;\leq\; l_{x1} \;\leq\; n_{\mathrm{mb},x1} - 2^{s+1}. \tag{6}$$

Any post-update policy violation is therefore a genuine bug rather than a criterion asking for too much. Second, an upstream AthenaK bug: the derefinement list was sorted with end iterator `&cllderef[ctnd-1]`, which *excludes the last element.* When a mixed-level derefinement left a fine-level parent in the unsorted last slot, coarser parents merging before it were vetoed by the tree's finer-neighbor check order-dependently — some parents of an annulus merged and some did not, splitting what must be a ring-complete derefinement (the policy check caught it as a partial ring, e.g. 4 of 8 members). Outside the shearing box the bug merely delays some derefinements by one AMR interval, which is why it went unnoticed upstream.

5.8.1 Dynamic ring creation and destruction: the moving-ring criterion

The dynamic exercise uses a deterministic user AMR criterion (`MovingRingRefine` in the `shwave` pgen): refine every MeshBlock whose x_1 extent overlaps $[x_c(t) - w,\, x_c(t) + w]$ with $x_c(t) = x_{c0} + A\sin(ft)$, derefine all others. The criterion depends only on x_1, so it naturally flags complete rings; the ring-sync logic and the vetoes above do the rest. With A chosen so the interval sweeps past the (vetoed) boundary columns, *all* refined rings are destroyed near each turning point and re-created on the way back — one full create/destroy cycle per $2\pi/f$.

```
$ATH -i inputs/shearing_box/epicycle_amr_ring.athinput          # 20 orbits, 1 level
$ATH -i inputs/shearing_box/epicycle_amr_ring2.athinput         # 2 orbits, 2 levels
$ATH -i inputs/shearing_box/epicycle_amr_ring_nofargo.athinput  # 2 orbits, non-FARGO
```

The 20-orbit epicycle run ($x_c = 7\sin t$, $w = 1.9$, one sweep per orbit) creates 2296 and destroys 2240 MeshBlocks (Fig. 3, left); the counts and all history columns are bit-reproducible across reruns. Mass drift is exactly zero. The epicyclic amplitude invariant of §5.7 drifts by +0.015% over 20 orbits, between the static ring (+0.008%) and the uniform grid (+0.064%), and the time-matched x-KE history differs from the *static*-ring run by at most 2.6×10^{-3} of the peak — an order of magnitude closer than either run is to the uniform grid (2.4×10^{-2}): the dynamic regridding adds essentially nothing beyond the static refined-boundary truncation already characterized in §5.7. The two-level variant (nested moving rings, 16−296 blocks, graded per Eq. (6)) drifts by 0 to six digits over 2 orbits; the non-FARGO variant exercises the momentum-offset boundary path with exact mass conservation. Two-panel movies with per-frame mesh outlines (`dens_compare_epi_unif_vs_epi_amr{,2}.mp4`) show the rings sweeping, collapsing, and re-forming with no visible artifacts.

5.8.2 Wave structure through a moving refinement boundary

The epicycle is spatially uniform, so it never exercises prolongation/restriction on real structure. The vortical-shwave AMR test (`hydro_incompress_shwave_amr_ring.athinput`) runs the JG05 problem of §5.1 under a moving ring ($x_c = 0.2 \sin t$, $w = 0.05$), so wave structure is prolonged, restricted, and shear-remapped across every regrid event, including full ring destruction near the turning points at 64^2:

resolution	peak x-KE err. (static ring)	peak x-KE err. (moving ring)	AMR vs. static ring	mass drift
64^2	−4.05%	−8.55%	4.7×10^{-2}	0.0
128^2	−0.29%	−0.27%	1.8×10^{-4}	0.0

(errors vs. the same-resolution uniform run; curve maxima relative to the peak x-KE; the 128^2 runs use CLI overrides `mesh/nx1=128 mesh/nx2=128`). This test is in fact the code-verification setup of Johnson & Gammie (2005, ApJ 635, 149, their Fig. 1, left): the vortical-shwave amplitude obeys their eq. (9),

$$\delta v_x(t) = \delta v_{x0} \, \frac{1 + \tau_0^2}{1 + \tau^2}, \qquad \tau = q\Omega t + k_{x0}/k_y, \tag{7}$$

with $\tau_0 = n_{wx}/n_{wy} = -4$ here, so the swing through $\tau = 0$ at $\Omega t = 8/3$ amplifies δv_x by $1 + \tau_0^2 = 17$. At 128^2 all three runs track Eq. (7) to better than 0.5% of the peak amplitude (uniform 0.39%, static ring 0.40%, moving-ring AMR 0.44%). At 64^2 the moving ring roughly doubles the static-ring truncation — there the ring covers only ~ 1.6 wavelengths of the $|n_{wx}| = 8$ mode and full ring destruction repeatedly restricts fine data mid-swing. At 128^2 the moving-ring and static-ring solutions are indistinguishable (1.8×10^{-4}, Fig. 3, right): the regridding cost converges away *at least* as fast as the static refined-boundary error, with no floor. (The apparent order is inflated by geometry — at 128^2 the same physical criterion keeps a relatively tighter ring that never fully vanishes — so this is a convergence statement, not an order measurement.)

5.9 Summary

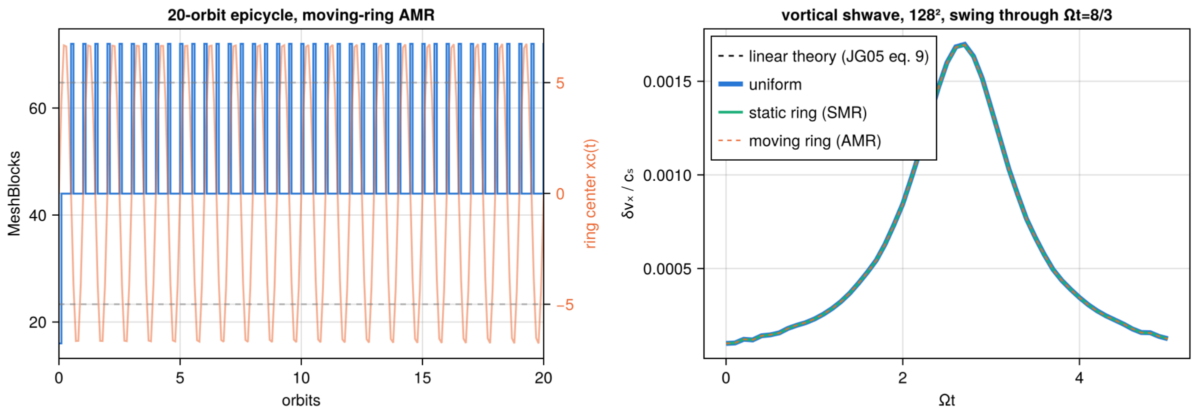

Figure 3: Phase 0c. Left: MeshBlock count (blue, left axis) in the 20-orbit moving-ring epicycle AMR run, with the prescribed ring center $x_c(t) = 7 \sin t$ (orange, right axis; dashed lines mark the vetoed boundary columns beyond $|x_1| = 5$). Rings are destroyed near every turning point and re-created on the way back, 2296 blocks created / 2240 destroyed in total. Right: radial-velocity perturbation amplitude $\delta v_x/c_s$ of the 128^2 vortical shwave through the swing, with the linear-theory amplitude of Eq. (7) (Johnson & Gammie 2005, eq. 9) overplotted — uniform, static SMR ring, and moving-ring AMR are visually indistinguishable from each other (moving vs. static ring: 1.8×10^{-4} of peak) and track the theory to $< 0.5\%$ of the peak.

Check	Metric	Result
SMR no-FARGO vs `test/shwave-smr` reference	all history columns	bit-identical
Uniform: FARGO vs non-FARGO	mass / x-KE	exact / 3.3%
Swing + gravity, non-FARGO vs linear theory	$L_2(\delta)$ / phase	1.2×10^{-2} / 10^{-12}
Non-FARGO cost (L_x=2π box)	cycles vs FARGO	5.7×, as predicted
Ring + FARGO: mass conservation	max drift	exact (0.0)
Shwave: ring-FARGO vs ring-non-FARGO	x-KE	3.8%
Epicycle, 20 orbits, ring + FARGO	amplitude drift / invariant	0.008% / flat to 10^{-4}
Epicycle, 20 orbits, boundary annuli	vs ring run, all columns	bit-identical (0.0)
Two-level (1+2) rings / bndry annuli	epicycle vs analytic; ring2 vs bndry2	5×10^{-6}; bit-identical
Two-level shwave (both configs)	mass / x-KE vs uniform	exact / 4.5–5.2%
Boundary annuli + FARGO: epicycle	max rel. vs analytic	5×10^{-6} (= uniform)
Boundary annuli + FARGO: shwave	mass / x-KE vs uniform	exact / 4.9%
Mixed-level x_1 boundary	—	fatal, names the level span
Partial-x_2 ring with FARGO	—	fatal, names the ring
AMR epicycle, 20 orbits (2296 MBs created)	mass / invariant drift	exact / +0.015%
AMR epicycle vs static ring	time-matched x-KE / peak	2.6×10^{-3} (vs 2.4×10^{-2} to unifo
AMR epicycle, two-level nested rings	amp. drift, 2 orbits	0 to six digits
AMR epicycle, non-FARGO	mass	exact
AMR shwave vs static ring, $64^2 \rightarrow 128^2$	curve max rel. diff	$4.7 \times 10^{-2} \rightarrow 1.8 \times 10^{-4}$
AMR determinism	reruns, all counts/columns	bit-reproducible

6 Phase 1: shear-periodic multigrid boundary conditions

6.1 Method: an identification, not an operator change

Shear-periodicity identifies the two radial faces with a time-dependent azimuthal offset,

$$u(x,\,y,\,z)\;=\;u(x+L_x,\;y-q\Omega L_x\,t_s,\;z),\qquad t_s=t \bmod T_{\rm shear},\quad T_{\rm shear}=\frac{L_y}{q\Omega L_x},\tag{8}$$

and nothing more: the interior 7-point Laplacian, the red–black smoother, restriction, and prolongation are all untouched. Only the x_1 ghost fill changes — the ghost column is the opposite boundary column with its content shifted by $\Delta y=\pm\,q\Omega t_s L_x$ (henceforth $\pm\,\mathrm{qomt}\,L_x$; the sign convention, validated in the FFT track, is: inner ghosts shift by $+$, outer by $-$). The shift is generally a non-integer number of cells *at every multigrid level*: the same Δy is 35.52 fine cells but 4.44 cells four levels down. The integer part is index arithmetic with periodic wrap; the fractional part ϵ is applied with the same conservative remap kernels (`DC/PLM/PPMX_RemapFlx`) shared by orbital advection, the hydro shear boundary, and the FFT solver (order selected by the new `<gravity> mg_remap`, default `plm`).

Two structural facts make this cheap. First, the fixed point of the V-cycle (and of FAS) satisfies the *finest-level* discrete system, so the remap quality on coarse levels affects only the convergence *rate*, never the converged answer — one code path serves every level with no accuracy tax. Second, because each boundary plane is remapped as a full-period global row, the wrap supplies all guard cells and the `ngh=1` multigrid ghost width imposes no stencil restriction even for PPMX.

The shear phase is frozen once per solve as $\mathrm{qomt}=q\Omega\,(t \bmod T_{\rm shear})$ evaluated at `pmesh->time` — the FFT solver's convention exactly, so both solvers see identical boundary identifications at any instant (hydro's shear BC uses $t+\mathrm{d}t$ on the final RK stage; that $O(\mathrm{d}t)$ offset is a pre-existing, FFT-validated convention).

6.2 The global boundary-plane algorithm

At each MeshBlock level the driver keeps two global boundary planes (full y–z extent of that level, `ngh` deep in x). One fill is three stream-ordered kernels: *gather* the interior columns adjacent to each x_1 face of the boundary blocks into the planes; *remap* each plane row by $\pm\,\mathrm{qomt}\,L_x$ in place; *scatter* into the x_1 ghost slabs of the opposite-face blocks — over the *entire* slab $k,j\in[0,n{+}2\,\mathtt{ngh})$, with periodic wrap in y and z supplying the slab's edge and corner ghosts (trilinear prolongation reads the full 3^3 stencil, so corners matter). The fill is invoked at the top of `MultigridDriver::PhysicalBoundary`, which already runs after every boundary exchange in all three MG task lists including the final finest-level exchange — so the one ghost layer of Φ handed to the `SelfGravity` source term is also sheared, and no task-graph surgery was needed. It overwrites the unsheared periodic copy the neighbor communication just wrote (the tree wraps `shear_periodic` x_1 exactly like `periodic`); at $\mathrm{qomt}=0$ the fill is skipped since that copy is already exact.

The root grid is a single replicated array, so `MGRootBoundary` gains a shear branch (`RootShearBoundaryX1`) that remaps the interior-row ghost columns in place; the existing x_2/x_3 passes then wrap the corner rows, and the 1-cell block level inherits correct ghosts through `SetFromRootGrid` unchanged. Mean-source subtraction — required for solvability — is re-enabled whenever every face is `periodic` or `shear_periodic` (it was silently disabled for any not-strictly-periodic mesh before, one of three defects that made shearing-box multigrid gravity *silently wrong* rather than unsupported: the ghosts were also overwritten with a $\Phi=0$ Dirichlet wall after the periodic copy).

Multi-rank support is one collective at one seam: each rank gathers only its own boundary blocks' columns into a zero-filled plane, and a single `MPI_Allreduce(SUM)` then reconstructs the global plane on every rank before the (rank-redundant) remap. Every plane cell has exactly one writer, so summing the zeros is exact and independent of reduction order — the multi-rank solve is *bit-identical* to the serial one. The root-grid fill needs no communication (the root grid is already replicated on every rank). Remaining scope is enforced by fatal guards (uniform grid, periodic x_2/x_3, no `mg_bc` override, `mg_nghost=1`, device root grid).

A byproduct worth noting: MG task lists are now executed through a driver-independent runner, so `pgrav->Solve(nullptr, 1)` from a problem generator works with `solver=multigrid` — static Poisson tests (below) run the multigrid solver through the identical code path as evolution.

6.3 Tests and results

All tests use the shared inputs of the FFT track with the single override `gravity/solver=multigrid` (the inputs now carry the MG parameters: `threshold=0.0` stagnation mode, FMG with one V-cycle per level, `mg_remap=ppmx`); the FFT solver runs the same input as the oracle. Multigrid requires cubic MeshBlocks, so the swing runs override `meshblock/nx*=16`.

Shear-periodic instants (plumbing arbiter). `fft_poisson_shear.athinput` with `solver=multigrid` at $t_0 = 0$ *and* at $t_0 = T_{\text{shear}} = 2/3$ (exercising the phase wrap) produces Φ **bit-identical** to the same solve with plain-periodic x_1 and no shearing box. Any plumbing or indexing error shows up here first.

Generic phase: residual, and what it does and does not prove. At $t_0 = 0.37$ ($\Delta y = 0.555\,L_y$) the discrete residual $L[\Phi]-4\pi G(\rho-\bar{\rho})$ evaluated with the solver-filled ghosts is 5.9×10^{-15} / 5.5×10^{-14} / 2.7×10^{-13} at $32^3/64^3/128^3$: the V-cycles converge to machine precision with no floor from the sheared boundary. But for multigrid this residual is a *self-consistency* check only — a converged solve with a wrong boundary identification would also drive it to zero (the FFT solver's $\sim 10^{-3}$ values on the same metric measure its remap truncation against the *exact* identification instead). The accuracy arbiter is the next test.

Analytic shearing wave (the sign arbiter). `profile=shwave` initializes a single rolled-frame Fourier mode whose discrete potential is known in closed form; the pointwise Φ error is $O(1)$ under any sign or offset error. Both solvers converge; the multigrid errors are *smaller* than the FFT solver's own, because MG remaps only the boundary ghosts while the FFT rolls and unrolls the entire field:

max rel. Φ error	32^3 (ϵ=0.76)	64^3 (ϵ=0.52)	128^3 (ϵ=0.04)
multigrid (`mg_remap=ppmx`)	3.2×10^{-5}	4.9×10^{-6}	9.5×10^{-8}
FFT (`fft_remap=ppmx`)	1.1×10^{-4}	1.3×10^{-5}	1.6×10^{-6}

(ϵ = fractional part of the cell shift at that resolution; it varies with N, so these are not clean order measurements — the FFT track's fixed-ϵ study established the remap orders.)

V-cycle convergence rate (the named open risk). Iterative mode (`full_multigrid=false`, warm start off a zero guess) at 64^3, $t_0 = 0.37$ — a near-worst fractional shift $\epsilon = 0.52$ — against the identical solve at shear phase zero:

	contraction/cycle	cycles to floor	floor
periodic phase (qomt = 0)	0.116	15	1.28×10^{-11}
sheared ($\epsilon = 0.52$)	0.157	17	1.27×10^{-11}

The remapped boundary costs two extra V-cycles and reaches the *same* roundoff floor — no stall, no loss of the solution. The open risk is retired.

Dynamic oracle A/B: swing amplification. swing_selfgrav.athinput (the Phase-0/FFT-track configuration: $128^2 \times 16$, self-gravity at $4\pi G = 1.9$, two swing times) run twice, changing only the solver. Peak amplification -3.511418 in **both** runs (relative difference 5.7×10^{-8}); the full 401-point amplification histories differ by at most 2.0×10^{-7} (rms 8.6×10^{-8}); phase locking (sine projection) holds at 4×10^{-14} (MG) vs. 8×10^{-14} (FFT). (§10.4 derives why the $d_{\cos}$ projection equals the wave amplitude exactly and why the sine channel is empty.) At matched settings the two solvers are interchangeable.

Multi-rank (the Allreduce seam). With -D Athena_ENABLE_MPI=ON, the static tests across 1–8 ranks reproduce the serial results: residuals at machine precision, and the shwave error *identical to all printed digits* (the plane reconstruction is exact, so the multi-rank solve is bit-identical to serial); the qomt = 0 shear-vs-periodic bit-identity also holds at 2 ranks. The 2-rank swing run matches the serial multigrid history to 10^{-11} — the MPI summation order of the history reductions, not the shear fill. (The rank-local diagnostics of the fft_poisson pgen were made MPI-aware in the process.)

Periodic regression. With no <shearing_box> block the code path is unchanged: the Jeans-wave evolution (jeans_wave vs. jeans_fft) reproduces the documented post-merge comparison — final density bit-identical between MG and FFT, velocities within 4×10^{-14} — and the static periodic Poisson test now also runs under solver=multigrid (residual 5.9×10^{-14}).

6.4 Consistency with the published Athena++ solver (Tomida & Stone 2023)

Before Phase 2 builds on this solver, we anchor it to the published results of the Athena++ multigrid method paper it descends from (Tomida & Stone 2023, ApJS 266, 7; docs/Tomida_2023_ApJS_266_7.pdf), reproducing their uniform-grid tests of §4.1 and §4.3 on our tree (notebook §10–11; inputs tests/ts41_sin3.athinpu tests/ts43_jeans.athinput). Their §4.2 (binary potential on refined meshes) is the planned acceptance test for Phase 2.

§4.1: static sinusoidal waves. Triple-sine density ($A = 1$, $\rho_0 = 2$, $G = 1$) in a fully periodic unit box, with the analytic potential of their eq. (29). The fft_poisson pgen gains profile=sin3 and a per-iteration study mode (<problem> conv_niter): in MGI mode successive Solve() calls warm-start from pgrav->phi, so N calls with niteration=1 reproduce N V-cycles from the zero initial guess, and ϵ (vs. analytic, their eq. 6), δ (vs. the fully converged discrete solution, eq. 5, built by a bit-identical reference pass so $\delta_{20} \equiv 0$), and the defect d (eq. 7) are measured after every iteration inside one process; in FMG mode iteration 1 is one full FMG sweep. Results at $h = 1/16 \ldots 1/256$ (their 1/512–1/1024 points exceed laptop memory):

- convergence factor per V-cycle 0.125 at 64^3 (rising to 0.127 at 256^3, 0.088 at 16^3) in both δ and d, for both MGI and FMG — their reported $\sim$0.125 (Yavneh 1996). The 0.116 measured in §6.3 on the modes+blob source is source-dependent variation, not a discrepancy;
- converged ϵ exactly second order (order 2.008 $\to$ 2.000), **identical to the FFT solver's ϵ to every printed digit** at all resolutions (both solve the same discrete system);

- the first FMG sweep alone already *below* the converged discretization error ($\epsilon_{\rm sweep}/\epsilon_{\rm conv} = 0.39$–$0.60$), their "remarkable" finding.

§4.3: Jeans-wave dynamical test. Periodic $1 \times 0.5 \times 0.5$ box; the `jeans_wave` pgen's oblique-wave machinery reproduces their $\mathbf{k} = 2\pi(1/L_x, 1/L_y, 1/L_z)$ exactly ($\lambda = 1/3$), and its `n_jeans` is their $\nu_{\rm Jeans}$ ($\lambda_J = \sqrt{\pi c_s^2/(G\rho_0)}$; stable case $\nu = 0.5$, $A = 10^{-6}$, $\Gamma = 5/3$, PLM + HLLC + `rk2` in place of their VL2). Two additions were needed: their *isothermal* pressure perturbation (`<problem> perturb=isothermal`; $\delta p = p_0 A \sin kx$), and their error metric (`ts_errors=true`): the rms of the L1 errors of the five conserved variables against the analytic linear solution. The isothermal perturbation is not an eigenmode — it excites a *static* self-gravitating mode ($\delta p = c_s^2\nu^2\delta\rho$ balancing gravity, fraction $f_{\rm st} = (1-1/\Gamma)/(1-\nu^2) = 8/15$ of the density perturbation) plus a standing gravito-acoustic wave at the dispersion-relation frequency — and the metric compares against this two-mode solution. (Its correctness is validated by the convergence itself: a wrong decomposition floors the error at $\mathcal{O}(10^{-7})$.) At $t = 1.5$ (the time of their Fig. 11; Fig. 10's is unstated), $N_x = 16 \ldots 128$:

rms of L1 errors	N_x=16	32	64	128
fully converged	3.60×10^{-7}	1.92×10^{-7}	3.98×10^{-8}	8.12×10^{-9}
one FMG sweep ($N_{\rm iter}$=0)	3.99×10^{-5}	6.00×10^{-6}	1.37×10^{-6}	3.13×10^{-7}
FFT (exact potential)	3.60×10^{-7}	1.92×10^{-7}	3.98×10^{-8}	7.86×10^{-9}

Both of their findings reproduce: the converged runs are roughly second order with a slight deviation (order 2.27–2.29; N_x=16 is pre-asymptotic at $\sim$5 cells per wavelength), while the single-sweep error is *precisely* second order (2.13) but 30–40× larger — gravity-dominated, so "use of the fully converged solutions improves the accuracy." The FFT solver (exact discrete potential every stage) bounds the converged curve to six digits.

6.5 Summary

Check	Metric	Result
Shear-periodic instants ($t_0 = 0$ and $T_{\rm shear}$)	Φ vs. periodic solve	bit-identical
Generic phase 32^3–128^3	discrete residual	6×10^{-15}–3×10^{-13}
Analytic shwave, 128^3	max rel. Φ error (MG / FFT)	9.5×10^{-8} / 1.6×10^{-6}
V-cycle contraction, $\epsilon = 0.52$	per-cycle factor (vs. periodic)	0.157 (vs. 0.116)
Swing amplification vs. FFT	peak / trajectory / phase	5.7×10^{-8} / 2×10^{-7} / 10^{-14}
Multi-rank, 1–8 ranks	shwave digits / qomt=0 identity	identical / bit-identical
Jeans periodic regression	final density MG vs. FFT	bit-identical
T&S §4.1: CF / ϵ order / FMG sweep	vs. published	0.125 / 2.00 / sub-truncation
T&S §4.1: converged ϵ vs. FFT	all N, all digits	identical
T&S §4.3: converged / one-sweep orders	vs. published	2.3 / 2.13, sweep 30–40× larger

Next: Phase 2 — multigrid gravity on refined shearing-box meshes; delivered for interior rings in §7.

7 Phase 2a: multigrid gravity on refined shearing-box meshes — interior rings

7.1 Scope and the incremental decomposition

Phase 2 combines the two halves validated so far — refinement in the shearing box (Phase 0) and the shear-aware multigrid solver (Phase 1) — in three steps that track *where refined blocks sit relative to the shear boundary*:

- **2a (this section): interior rings.** Under the Phase-0 interior-x_1 policy, refined blocks never touch the shear-periodic x_1 faces, so the boundary blocks — the only participants in the Phase-1 shear-plane fill — all remain at the root level. No octet ever needs a sheared ghost.
- **2b: boundary annuli.** Uniformly refined x_1 boundary faces (allowed by the Phase-0c-lite hydro policy) put octets *at* the shear faces and need a sheared octet ghost fill — the host-side analogue of §6.2.
- **2c: AMR.** The shear planes and offset tables are sized once at construction; adaptive refinement adds a rebuild-after-remesh step (the analogue of `ShearingBox::ReinitAfterMeshUpdate`).

The `MGGravityDriver` constructor now enforces exactly this decomposition: with shear-periodic x_1 it aborts if refinement is *adaptive* (2c), or if any x_1 boundary MeshBlock sits above the root level (2b) — a check stricter than the hydro-side `CheckShearingBoxRefinement`, which also admits uniformly refined boundary faces. Static interior rings pass.

7.2 Step 0: the binary-potential octet control (no shear)

Upstream #776 ships the Athena++ binary-potential test (`pgen/tests/binary_gravity.cpp`, `tests/binary_grav` — precisely the mesh-refinement acceptance test of Tomida & Stone §4.2: two uniform spheres (M_1=2 at x=6/1024, M_2=1 at $x = -12/1024$, $R = 6/1024$, 10^3-subcell surface averaging) under four nested static refinement levels. Since none of our Phase-0/1 work had ever exercised the octet machinery (every earlier multigrid run was uniform), we run this control *first*, so that any later failure on sheared refined meshes indicts the shear coupling and not the inherited octet code.

Two caveats about the shipped input. First, its periodic boundaries are *not comparable* to the isolated two-sphere analytic solution the pgen's error norms use (the errors come out $O(1)$ by construction); the meaningful configuration is the paper's: outflow mesh boundaries with `gravity/mg_bc=multipole`, `mporder=4` (hexadecapole), now present as overridable defaults in the input. Second, the pgen's printed "Potential L2" is $\sqrt{\langle|\delta\Phi/\Phi|\rangle}$ (Athena++ convention), and the `<refined_region>` blocks are *absolute physical* volumes: doubling the root resolution leaves every region at its absolute size, so between the 64^3- and 128^3-root runs only the innermost box gains resolution and the global mean is frozen *by construction* — region-wise norms are the meaningful metric.

With 40 V-cycles (FMG), roots $32^3/64^3/128^3$ + 4 levels (232/232/288 blocks; at 64^3 the level-1 region force-refines all root blocks, so the far field lands at h=1/128 in both larger runs):

rms rel. Φ error per region	R_0 (far)	R_1	R_2	R_3	R_4 (inner)
32^3 root	1.26×10^{-4}	1.58×10^{-4}	2.56×10^{-4}	3.43×10^{-4}	9.57×10^{-4}
64^3 root	3.19×10^{-5}	4.23×10^{-5}	6.55×10^{-5}	8.68×10^{-5}	2.44×10^{-4}
ratio	3.94	3.73	3.91	3.95	3.92

Second order in every region (the h of each region halves exactly between these two runs). From 64^3 to 128^3 only R_4 refines ($h = 1/1024 \to 1/2048$) and its error drops by 2.37 — saturated by the unimproved surrounding regions, whose inherited error it cannot undercut; the same-resolution regions agree between the two runs to four digits (a strong bit-level reproducibility check through the float32 snapshots). On the Tomida & Stone normalized-rms metric (their eq. 31) the 64^3 run gives 7.5×10^{-5} against their Fig. 7(d) value of $\sim 1.3 \times 10^{-4}$ at h=1/64 — consistent, ours slightly lower because of the force-refined far field. The late-stage convergence factor is $\sim$0.5 per V-cycle, their reported refined-mesh value (§6.3 measured 0.12–0.16 on uniform grids: refinement, not shear, is what costs convergence rate). The defect reaches its 10^{-12} relative floor. The octet machinery works as published.

7.3 What 2a required — and what it did not

The entire code change is the constructor guard described above. The Phase-1 shear-plane fill needed *nothing*:

- `FillShearGhostPack` gathers and scatters only blocks whose own `mb_bcs` carry `shear_periodic` on an x_1 face. Interior fine blocks have internal (`BoundaryFlag::block`) faces there, so they are skipped automatically — their (root-unit, hence meaningless) entries in the per-block offset table are never read. No level test was needed.

- The plane geometry (`nmmbx2_` $\times$ block cells per MG level) is expressed in root-grid units, and under the 2a guard every participating block *is* a root-level block, so the planes tile exactly as on the uniform grid.

- Octet x_1 neighbor tables already wrap shear-periodic like periodic (Phase-1 groundwork); under the 2a guard no octet abuts the x_1 faces, so the unsheared wrap is never consulted. (It becomes live — and wrong — only in Phase 2b, which is why boundary-level refinement stays fatal.)

SMR additionally requires `nghost=4` (AthenaK wants even ghosts with refinement) and, as always for this solver, cubic MeshBlocks; the test inputs use 16^3 blocks, giving four x_1 block columns of which the middle two are refined.

7.4 Static tests: bit-identity, the analytic arbiter on SMR, MPI

Input `tests/mg_poisson_shear_smr.athinput`: the Phase-1 shear test (§6.3) with a level-1 ring over $x_1 \in [-1/4, 1/4]$ (full x_2/x_3).

Shear-periodic instants. At $t_0 = 0$ the refined-mesh solve with shear-periodic x_1 is **bit-identical** (float64 comparison of every cell of every block) to the same mesh with plain-periodic x_1, for both the multi-mode and shwave sources. (Compare the bin *data*: the file header echoes the input, so whole-file comparison always differs.)

Analytic shearing wave. The reference potential $\Phi_{\rm ref}$ is the *exact solution of the discrete system on the uniform root grid* — not the continuum potential. The shwave profile initializes a single rolled-frame Fourier mode,

$$\rho = \rho_0 + A\cos(k_x x + k_y y + k_z z), \qquad k_x = \frac{2\pi}{L_x}\Big(n_1 + \mathrm{qomt}\,\frac{L_x}{L_y}\,n_2\Big), \quad k_y = \frac{2\pi}{L_y}n_2, \quad k_z = \frac{2\pi}{L_z}n_3, \tag{9}$$

a plane wave tilted by the current shear phase and therefore exactly shear-periodic across the x_1 wrap. Acting on a cosine, the 7-point Laplacian has the exact (no-expansion) eigenvalue

$$\lambda_h = \frac{2\cos(k_x\Delta x) - 2}{\Delta x^2} + \frac{2\cos(k_y\Delta y) - 2}{\Delta y^2} + \frac{2\cos(k_z\Delta z) - 2}{\Delta z^2}, \tag{10}$$

so $L_h\Phi = 4\pi G(\rho - \bar{\rho})$ is solved exactly by $\Phi_{\rm ref} = [4\pi GA/\lambda_h]\cos(k_x x + k_y y + k_z z)$ at cell centers, with the Δ's at the *root* spacing and both fields mean-subtracted (the periodic gauge). This is the closed form behind the pgen's `SHWAVE ERROR` line, inherited from the FFT track — the rolled k-space kernel *is* $1/\lambda_h$, which is why FFT solves it to machine precision. The choice fixes the attribution: on a *uniform* grid $\Phi_{\rm ref}$ is exact, so a correct solver deviates only through its ghost-fill remap (Fig. 4, third panel: 5×10^{-6}, and the roundoff floor at qomt $= 0$); against the *continuum* $-1/k^2$ instead, both regions would show their own $O(h^2)$ discretization error ($\sim (k\Delta x)^2/12$ per direction, 8×10^{-4} at the root spacing) and the refinement structure would blur. On the refined mesh the composite operator is *not* the root-grid operator: the fine region's eigenvalue is closer to the continuum, differing per direction by $(kh)^2/12$ terms that sum to 6.0×10^{-4} relative at 64^3 — and the measured fine-region plateau is 6.7×10^{-4}: the fine grid deviating from $\Phi_{\rm ref}$ by *being more accurate than it*. The pointwise maximum sits in the first fine cell at the refinement interface ($x = \pm0.246$ at 64^3), symmetric about the box center, decaying harmonically into the coarse region, with *no* feature at the shear faces $x = \pm1/2$; at the generic phase it is within 3% of the qomt=0 value at 64^3 and 0.6% at 128^3 — the interface, not the shear, owns the error:

	qomt $= 0$		generic phase $t_0 = 0.37$	
	max rel.	L_2 rel.	max rel.	L_2 rel.
32^3	1.18×10^{-2}	5.05×10^{-3}	1.44×10^{-2}	8.14×10^{-3}
64^3	5.51×10^{-3}	1.28×10^{-3}	5.66×10^{-3}	2.06×10^{-3}
128^3	2.66×10^{-3}	3.25×10^{-4}	2.68×10^{-3}	5.18×10^{-4}
L_2 ratios		3.94, 3.94		3.96, 3.97

L_2 is cleanly second order at both phases — the same interface behavior Tomida & Stone report (their rms metric) — while the interface *peak* converges at first order, the known local property of the mass-conserving level-boundary stencil. The shear remap contributes only the small, second-order-converging difference between the two L_2 columns. Figure 4 shows all of this at once: the error concentrates in the fine region (the reference uses the root-grid eigenvalue) and peaks on the interfaces, while the shear faces are featureless — and the same metric on the uniform Phase-1 run is three decades smaller, so the entire visible structure is refinement, not shear. One warning: the `profile=modes` residual norms print $O(1)$ on any refined mesh because the pgen evaluates a naive per-block Laplacian, which is *not* the composite multigrid operator at level interfaces; on SMR the shwave comparison is the valid metric.

Contraction and MPI. Static FMG solves contract at 0.18/cycle on this two-level mesh (identically for periodic and sheared boundaries; uniform grids gave 0.12–0.16). Runs at 1/2/4 ranks print identical doubles and their float32 snapshots agree to one ulp (5.8×10^{-11}, np=2 $\equiv$ np=4).

7.5 Dynamic test: swing amplification on a ring, with control

Input `tests/swing_selfgrav_smr.athinput`: the §6.3 swing configuration ($128^2{\times}16$, $4\pi G = 1.9$, two swing times) with a level-1 ring nominally over the middle half in x_1 — in fact over the middle *six* of eight block columns: the region edges ($\pm\pi/2$ as a decimal literal) sit 3×10^{-15} inside the

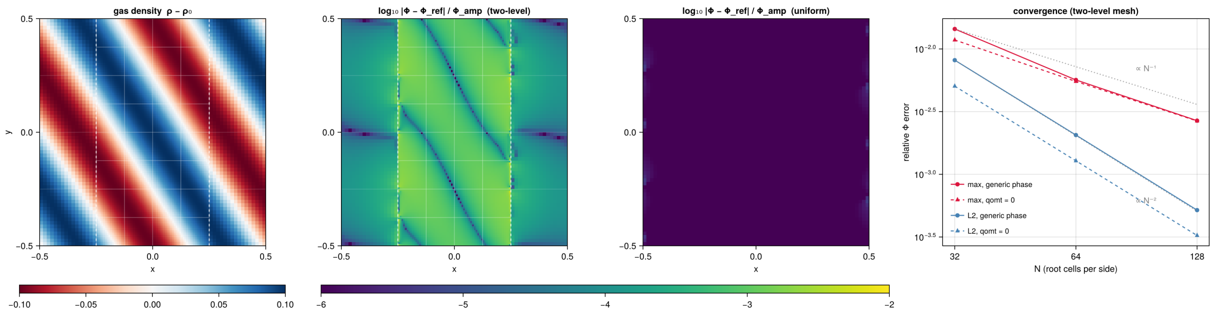

Figure 4: The static shwave test at the generic shear phase (64^3 root, level-1 ring over $|x| < 1/4$, block outlines in white). First panel: the sheared density mode. Second: $\log_{10}$ relative potential error against the root-grid closed form (eq. 10) on the two-level mesh — the error lives in the *fine* region (root–fine eigenvalue mismatch, the 6×10^{-4} plateau of §7.4), peaks in the first fine-cell layer on the refinement interfaces (dashed; bright cells modulated by the mode phase, pinched at its zero-crossing curves), and shows no feature at the shear-periodic faces $x = \pm 1/2$. Third: the *same* metric on the uniform Phase-1 run — pure remap truncation at 5×10^{-6}, three decades below the interface and eigenvalue structure. Fourth: error vs. resolution at both shear phases — L_2 second order, interface peak first order, the two phases nearly coincident.

neighboring columns and pull them in, a wholesome warning about `<refined_region>` edges at non-representable coordinates; the uniform twin is the single override `mesh_refinement/refinement=none`, the oracle twin additionally `gravity/solver=fft`.

	peak $d_{\cos}/A$	peak time	max $\lvert d_{\sin}\rvert/A$
uniform, FFT	−3.511418	7.804	2.8×10^{-13}
uniform, multigrid	−3.511418	7.804	0.7×10^{-13}
ring SMR, multigrid	−3.583481	7.804	1.3×10^{-13}

The uniform twins reproduce the Phase-1 A/B exactly (the 16^3-block relayout and `nghost=4` change nothing). The SMR peak differs from the uniform one by 2.05% — *toward* the converged answer (linear theory 3.605; the 256^2 FFT-track run gave 3.62), as refining most of the domain should move it, at machine-level phase lock. The decisive attribution is the **gravity-off control** (`hydro_srcterms/self_gravity=false`, both meshes): the hydro-only SMR-vs-uniform deviation is *larger* than the self-gravitating one — peak shift +2.78% vs. +2.05%, normalized trajectory deviation 7.7×10^{-2} vs. 5.6×10^{-2} — so multigrid gravity on the refined shearing mesh contributes *no* deviation of its own beyond hydro truncation. Figure 5 shows the three amplification histories (indistinguishable at plot scale) with the pairwise differences: the SMR-vs-uniform gap rides at 10^{-2}–10^{-1} while the MG-vs-FFT solver seam sits five decades below it; Figure 6 compares the final density fields.

7.6 The stopping rule on refined meshes (practical note)

The first full-length SMR swing run stopped at $t = 5.5$: `threshold=0.0` (stagnation-exit mode) reached the driver's 40-V-cycle cap — *while still converging healthily* (defect 2.9×10^{-12}) — and the cap gracefully ends the run via `nlim`. The mechanism: near the swing peak the solution is dominated by the wound-up $k_x \simeq 0$, $k_y = 1$ mode, whose coarse-grid correction must pass through the sheared x_1 wrap where the y-remap acts on 8-cell rows; the warm-started contraction degrades

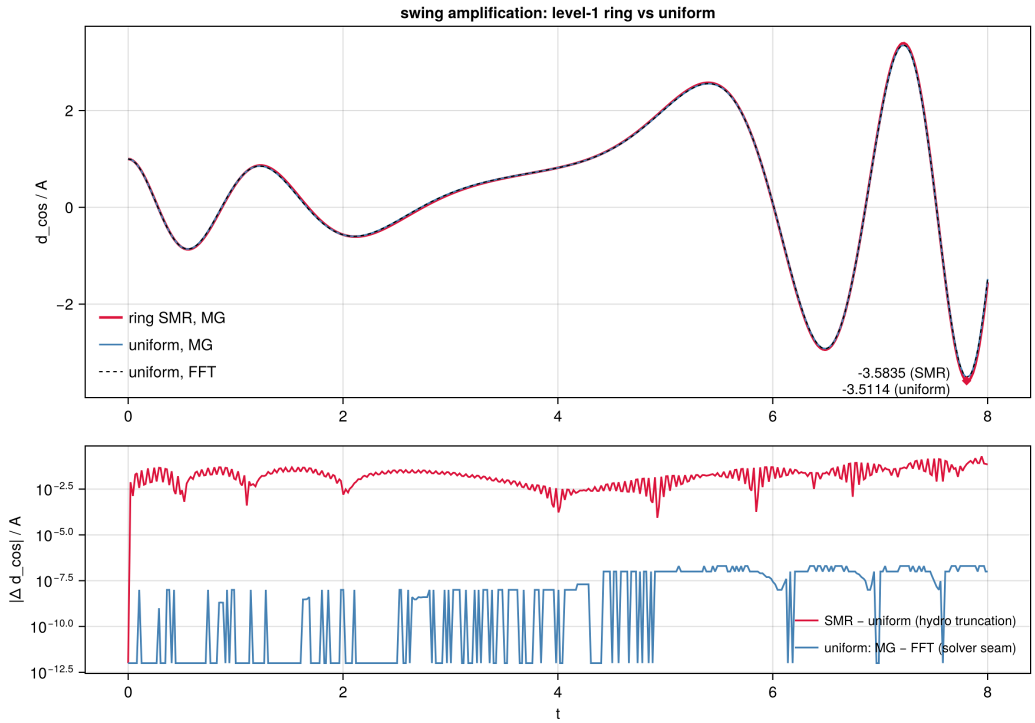

Figure 5: Swing amplification histories $d_{\cos}/A$ for the three runs (top; the curves overlap to the line width, peaks annotated) and the pairwise trajectory differences (bottom, log scale): SMR-vs-uniform is the hydro-truncation resolution blend; uniform MG-vs-FFT is the solver seam, floored by the history-file print precision.

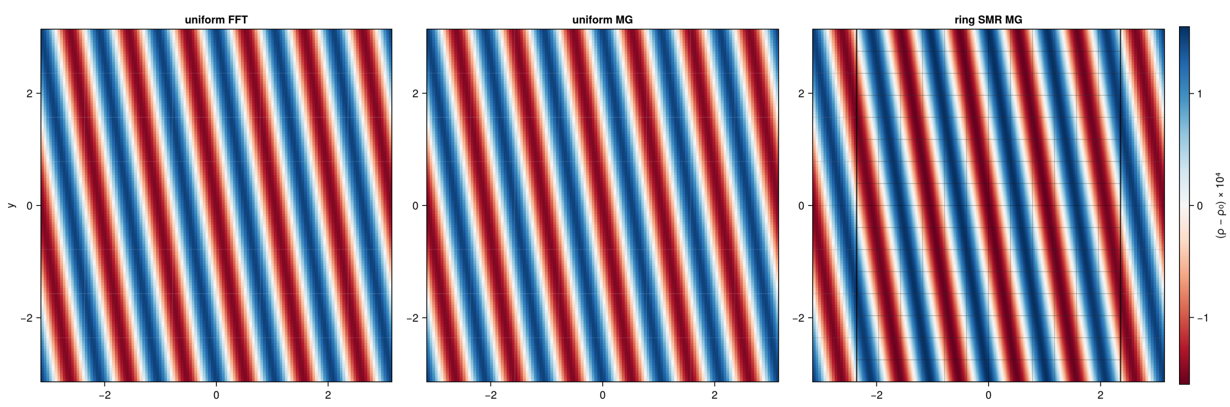

Figure 6: Final ($t = 8 = 2\,t_{\rm swing}$) density deviation $(\rho - \rho_0) \times 10^4$ at the z mid-plane for the three swing runs: uniform FFT, uniform multigrid, and the level-1 interior-ring SMR run (block outlines drawn; the heavier lines mark the refinement-level boundary). The two uniform panels are indistinguishable ($\leq 2 \times 10^{-7}$ everywhere); the SMR panel carries the same trailing wave with the amplitude difference of §7.5.

to $\sim$0.55/cycle (static solves on the same mesh: 0.19), so the defect ratio never exceeds the 0.9 stagnation trigger within 40 cycles. This is a stopping-rule interaction, not a solver failure — and it is exactly why Tomida & Stone run production refined-mesh gravity as FMG with a *fixed* iteration count. The SMR inputs therefore default to `threshold=-1`, `niteration=10`; on the uniform twin, fixed-10 and stagnation mode agree to 10^{-11}, and the full-length SMR run completes with no MG warnings.

7.7 Summary

Check	Metric	Result
Binary potential (octet control)	region-wise order $32^3 \to 64^3$	3.7–3.95
Binary potential vs. T&S Fig. 7(d)	$\epsilon_{\rm rms}$ at h=1/64 root	7.5×10^{-5} vs. $\sim 1.3 \times 10^{-4}$
SMR shear instants ($t_0 = 0$)	Φ vs. periodic twin	bit-identical
SMR analytic shwave	L_2 order (qomt=0 / 0.37)	3.94 / 3.96–3.97
SMR contraction (static)	per-cycle, periodic = sheared	0.18
Multi-rank 1/2/4	printed digits / f32 snapshots	identical / 1 ulp
Swing: uniform twins	peak, vs. Phase-1	-3.511418, exact
Swing: ring SMR	peak shift, direction	+2.05%, toward theory
Swing: gravity-off control	hydro-only shift (must dominate)	+2.78%
Swing: phase lock	$\max \lvert d_{\rm sin} \rvert / A$, all runs	$\sim 10^{-13}$

Next: Phase 2b — sheared octet ghosts, so refined blocks may sit *on* the shear boundary (boundary annuli); delivered in §8.

8 Phase 2b: sheared octet ghosts — refinement on the shear boundary

8.1 Where the shear enters the octet machinery

With boundary annuli — both x_1 faces uniformly refined over the full x_2/x_3 extent, the configuration Phase 0c-lite validated for hydro — the multigrid hierarchy places *octets* (per-cell 2^3 refinement units bridging the block levels down to the root grid) directly on the shear-periodic faces. Phase 1 had already wrapped the octet x_1 neighbor tables like periodic and exempted the shear faces from the physical-boundary reflection, so octet x_1 ghosts received *unsheared* periodic values — correct only at qomt = 0, and the reason 2b was fatal-guarded. Every consumer of those ghosts (octet smoothing, the FAS right-hand side, restriction, FMG and V-cycle prolongation, and the octet$\to$block transfer `SetFromRootGrid`) funnels through exactly two chokepoints, `SetBoundariesOctets` (per level, inside the V-cycle) and `SetOctetBoundariesBeforeTransfer` (before the root$\to$block transfer), which makes the fix local.

Two structural facts shape the implementation. First, octets are *host-side* flat buffers, and they are *globally replicated* on every rank (the block$\to$root transfer runs an `MPI_Allgatherv` and every rank populates every octet), so the sheared fill is plain serial host code with **no MPI seam at all**. Second, under the uniform-boundary-level policy the boundary octets tile the full y–z surface at every octet level — precisely the precondition for the Phase-1 global-plane algorithm to transplant.

8.2 The fill

`FillShearOctetGhosts(lev, folddata)` is the Phase-1 algorithm at octet granularity: gather the interior cell columns adjacent to each x_1 face from the boundary octets into a per-level (y, z) plane (level resolution 2 (nrbx2 $\ll$ lev) cells), remap each row by $\pm$ qomt L_x, and scatter into the x_1 ghost slabs of the opposite-face octets — the whole slab including its y/z edge and corner ghosts (trilinear and flux-conserving prolongation read those), overwriting the unsheared values the neighbor-table exchange just wrote, with the same sign convention as `FillShearGhostPack`. The conservative remap runs through new *scalar per-face twins* of the team kernels (`DC/PLM/PPMX_RemapFlxFace` in `remap_fluxes.hpp`, host-callable, kept in documented lockstep with the validated device versions — the device path is untouched, preserving every Phase-1/2a result bit-for-bit). As in Phase 1, octet ghost quality can only affect the convergence *rate*: the V-cycle fixed point is set by the finest block level, whose ghosts come from the (generalized) block planes.

On the block side, the only change is plane geometry: the participating blocks — the only ones whose `mb_bcs` carry `shear_periodic` — now sit at a uniform level L above root, so the planes are sized at that level (`shear_reflev_` $= L -$ root); the per-block offsets were already expressed at each block's own level. The constructor guard weakens accordingly: x_1-face blocks must share *one* uniform level (the hydro invariant), AMR remains fatal (Phase 2c), and the deferred host-root shear fill stays guarded.

8.3 Static validation

Input `tests/mg_poisson_shear_bndry.athinput`: the §7.4 problem with the level-1 ring replaced by two annuli hugging the x_1 faces ($|x| \in [1/4, 1/2]$, full x_2/x_3); a two-level variant (level-2 annuli of width 1/16 plus the automatically graded level-1 buffers) runs off the same input via CLI overrides. The Phase-2a interior-ring results reproduce digit-exact with the new code in place (the octet fill is inert without boundary octets).

- **Shear-periodic instants.** At $t_0 = 0$ the annuli solves are **bit-identical** to their periodic twins — at one *and* two refinement levels — and their norms equal the interior-ring values to all printed digits, as they must: the two meshes differ by a half-box shift in x, under which the periodic mode's norms are invariant.

- **Generic phase** ($t_0 = 0.37$): $L_2 = 2.057 \times 10^{-3}$, *equal to the interior-ring value to four digits* (the same half-shift symmetry, now with the remap active across the refined wrap); max $= 5.95 \times 10^{-3}$ at the annulus' interior interface — 5% above the ring's peak, and still no feature *on* the shear faces themselves, now covered by fine blocks. Two-level: 6.03×10^{-3} / 4.88×10^{-3} (the larger L_2 is the root-eigenvalue reference artifact growing with off-root volume, §7.4).

- **Contraction.** Static FMG: 0.18–0.20/cycle at both one and two levels, with the *sheared* runs reaching floors slightly below their periodic twins (2.0×10^{-10} vs. 8.2×10^{-10} at one level) — the sheared octet coarse correction costs nothing.

- **Multi-rank.** np = 2/4 reproduce the serial digits exactly (the octet fill is replicated host work; the block planes keep their Phase-1 Allreduce).

8.4 Dynamic validation: swing on boundary annuli

Input `tests/swing_selfgrav_bndry.athinput`: the §7.5 swing configuration with the level-1 interior ring replaced by two annuli hugging the x_1 faces (4 of 8 block columns refined — two per face, the same edge-representation effect as §7.5 — vs. the ring's 6). The full-length run completes with no multigrid warnings:

	peak $d_{\cos}/A$	shift vs. uniform	max $\lvert d_{\sin}\rvert/A$
uniform, multigrid	-3.511418	—	0.7×10^{-13}
annuli, multigrid	-3.550196	$+1.10\%$	1.4×10^{-13}
ring (§7.5)	-3.583481	$+2.05\%$	1.3×10^{-13}

All three peak at the same $t = 7.804$ at machine-level phase lock. The annuli shift is again *toward* the converged answer (theory 3.605), and its size scales with the refined volume — two-thirds of the ring's refined columns, about half its shift. The gravity-off control closes the attribution exactly as in §7.5: hydro-only annuli-vs-uniform deviates *more* than the self-gravitating run (peak $+1.70\%$ vs. $+1.10\%$; normalized trajectory deviation 6.4×10^{-2} vs. 4.3×10^{-2}) — the sheared octet machinery and level-1 shear planes add no deviation of their own beyond hydro truncation.

8.5 Summary

Check	Metric	Result
2a regression	interior-ring shwave digits	exact
Annuli shear instants, 1 and 2 levels	Φ vs. periodic twin	bit-identical
Annuli vs. ring, qomt = 0	norms (half-shift symmetry)	all digits
Annuli generic phase	L_2 vs. ring L_2	equal to 4 digits
Contraction, 1 and 2 levels	per-cycle, sheared vs. periodic	0.18–0.20, floor below
Multi-rank np = 2/4	printed digits	identical
Swing: annuli peak	shift vs. uniform, direction	$+1.10\%$, toward theory
Swing: gravity-off control	hydro-only shift (must dominate)	$+1.70\%$
Swing: phase lock	max $\lvert d_{\sin}\rvert/A$	1.4×10^{-13}

Next: Phase 2c — AMR in the shearing box with multigrid gravity; delivered in §9.

9 Phase 2c: adaptive refinement with multigrid gravity in the shearing box

9.1 What the remesh invalidates, and where it gets rebuilt

Phases 2a–2b sized every shear structure once, at construction: the per-MG-level block boundary planes and their per-block (y, z) offsets, the boundary-level factor `shear_reflev_`, and the per-octet-level face flags `oct_shear_face_`. Adaptive refinement invalidates all of them — blocks appear, disappear, and migrate between ranks; the octet population changes; in principle even the boundary level can change. The fix rides an existing vehicle: `MultigridDriver::PrepareForAMR` already detects any remesh (via the mesh's AMR/load-balance sequence counter) before every solve and rebuilds the block arrays and octets. Phase 2c makes `AllocateShearPlanes` idempotent (planes are `realloc`'d, offsets recomputed from the current logical locations — load balancing can reassign blocks without changing this rank's block count) and calls it from the same rebuild branch; the octet face flags were already recomputed inside `InitializeOctets`. The constructor's adaptive fatal is removed: the uniform-boundary-level invariant is re-enforced after *every* AMR update by the hydro-side `Mesh::CheckShearingBoxRefinement` (Phase 0c), whose graded-refinement cap also keeps 2:1 balancing from ever refining the boundary columns.

9.2 Dynamic validation: swing with a prescribed moving ring

The swing pgen gains the shwave pgen's prescribed moving-ring criterion (`amr_moving_ring`: refine MeshBlocks overlapping $[x_c(t) - h_w,\, x_c(t) + h_w]$, $x_c(t) = x_{\rm amp} \sin(\omega_r t)$, derefine all others); input `tests/swing_selfgrav_amr.athinput` sweeps a thin ring ($x_{\mathrm{amp}} = \pi/4$, $h_w = \pi/16$: one to two block columns) through the swing. The full-length run creates 392 and destroys 280 MeshBlocks — a remesh roughly every fifth cycle, each one exercising the shear-plane/octet rebuild — with no multigrid warnings:

	peak $d_{\cos}/A$	vs. static uniform	norm. traj. dev.
moving-ring AMR, gravity on	-3.510989	-0.01%	2.9×10^{-2}
moving-ring AMR, gravity off	$+1.7403$ (peak $\lvert d\rvert$)	$+0.38\%$	5.1×10^{-2}

The thin transient ring adds negligible net resolution, so the gravity-on peak lands on the uniform value; the gravity-off control again *dominates* both the peak shift and the trajectory deviation — several hundred remesh events add nothing beyond hydro regridding truncation.

One metric changes character under AMR and deserves its attribution: the phase lock max $\lvert d_{\sin}\rvert/A$ is 9.9×10^{-5} (gravity on) — nine decades above the 10^{-13} of every static-mesh run. This is *regridding* truncation, not a shear or gravity error: each remesh prolongates the wave onto newly created fine blocks, and second-order interpolation of a cosine injects a sine component at truncation level. The gravity-off control shows the same signature at 3.7×10^{-5} — the ratio matching the amplified wave's larger amplitude — so the sheared multigrid machinery preserves phase exactly as well as the hydro does.

9.3 The fixed-ring limit

With $x_{\mathrm{amp}} = 0$ and $h_w = \pi/2$ the criterion reproduces the static-SMR ring footprint and never remeshes after the initial refinement — but *bit*-identity with the §7.5 run is structurally impossible: user-criterion AMR starts on the unrefined root mesh and refines within the first `refinement_interval` cycles, so the first few steps run at the coarse CFL. After that transient the two runs are the same computation: the meshes are identical block-for-block, the final density fields agree to 4.8×10^{-7} (the float32 snapshot floor is 10^{-7}; the wave amplitude 3.5×10^{-4}), the peaks differ by 1.2×10^{-4} relative, and — decisively — the phase lock returns to 4×10^{-14}: with no mid-run remesh the machine-level $d_{\sin}$ of the static runs is recovered, closing the regridding attribution of §9.2 from the other side.

9.4 Multi-rank

The 2-rank moving-ring run reproduces the serial one *to every printed digit*: identical MeshBlock-cycle count (identical mesh evolution), identical amplification history (maximum trajectory difference exactly zero at history precision), identical peak and $d_{\sin}$. The remesh path rebuilds the planes and offsets on every rank from replicated mesh metadata, the octet fill is replicated host work, and the block-plane Allreduce is exact by construction (Phase 1) — so rank count drops out of the entire AMR + shear + gravity loop. This also exercises the hydro-side AMR shear-state rebuild (`ShearingBox::ReinitAfterMeshUpdate`) under MPI, which Phase 0c had written but never tested multi-rank.

9.5 Summary

Check	Metric	Result
Moving-ring AMR swing	MBs created/destroyed, MG warnings	392/280, none
Peak vs. static uniform	gravity on	−0.01%
Gravity-off control	peak shift / traj. dev. (must dominate)	+0.38% / 5.1×10^{-2}
Phase $d_{\rm sin}$ under AMR	regridding truncation (off-control ratio)	amplitude-scaled
Fixed-ring limit vs. static SMR	final ρ / peak / $d_{\rm sin}$	5×10^{-7} / 1.2×10^{-4} / 4×10^{-14}
Multi-rank np = 2 AMR	history vs. serial	identical, all digits

Next: §10 pushes the machinery into the gravitationally unstable regime; the Gammie (2001) gravitoturbulence program (cooling, Q/α diagnostics) follows.

10 Into the unstable regime: the swing-run compendium and first collapse

10.1 All swing runs at a glance

Every swing-amplification run of this program shares the base configuration: $128^2{\times}16$ cells (16^3 MeshBlocks), $L_x{=}L_y{=}2\pi$, $L_z{=}\pi/4$, isothermal $c_s{=}1$, $\Omega{=}1$, $q{=}3/2$ ($\kappa{=}1$), $\rho_0{=}1$, single leading wave $(n_{wx}, n_{wy}) = (-6, 1)$, FARGO on, multigrid gravity with FMG + `niteration=10` unless noted. The gravest box modes obey $\omega^2 = \kappa^2{+}c_s^2{-}4\pi G\rho_0 = 2{-}4\pi G$, so $4\pi G = 2$ is the box-scale Jeans threshold.

run	mesh	$4\pi G$	amp	$t_{\rm end}$	outcome	peak
validation ladder (linear, stable): §6.3, §7.5, §8.4, §9.2						
P1 uniform (MG ≡ FFT)	uniform	1.9	10^{-4}	8	swing	−3.511418
P2a ring SMR (+twins, off)	6/8-col ring	1.9	10^{-4}	8	swing	−3.583481
P2b annuli (+off)	4-col annuli	1.9	10^{-4}	8	swing	−3.550196
P2c AMR ring / fixed (+off, np2)	moving/static	1.9	10^{-4}	8	swing	−3.5110/−3.5839
unstable window (linear)						
window super-swing	uniform	2.1	10^{-4}	8	swing, $\omega^2{<}0$ in $\lvert k_x\rvert{<}0.32$	−4.4569
nonlinear ladder (max ρ/ρ_0 in the peak column)						
bounce I	uniform	3.0	0.05	10	crest bounces	1.37
bounce II	uniform	3.0	0.10	16	crest bounces	1.73
marginal miss	uniform	5.0	0.05	10	crest $\lambda_J{\approx}$ width	2.70
collapse	uniform	**6.0**	**0.20**	10	**bound clump**	**203**
collapse, AMR zoom	density rings	6.0	0.20	10	§10.3	§10.3

Figure 7 completes the linear grid: uniform and moving-ring AMR at each of $4\pi G = 1.9/2.1/3.0$ (amp $= 10^{-4}$) against the linearized shearing-sheet ODE. Peaks: $-3.51/-3.51$ (theory -3.61), $-4.46/|4.21|$ (theory -4.52), and $-8.53/-8.26$ (theory -8.02) — the amplification nearly doubles between 1.9 and 3.0 as the $\omega^2 < 0$ window widens, the AMR runs track their uniform twins to 0.1–0.4 in $d_{\rm cos}/A$ (hydro-truncation level), and at $4\pi G = 3.0$ the growing spread between simulation and continuum theory is the 128^2 resolution limit amplified through the window, with the AMR curve between the two.

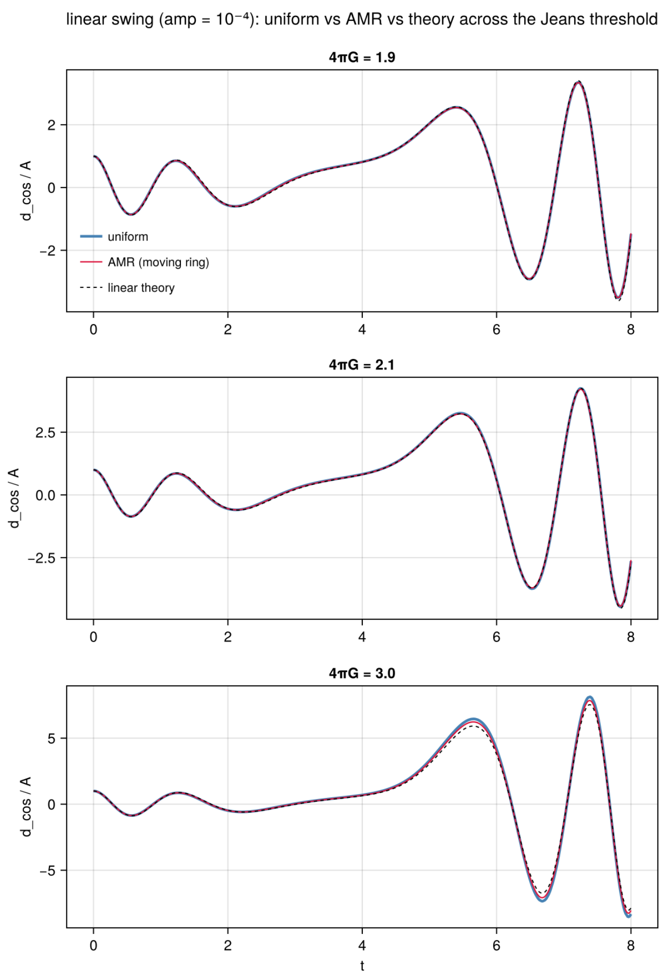

Figure 7: The linear swing grid: uniform (blue) vs. moving-ring AMR (red) vs. the linearized ODE (dashed) at $4\pi G = 1.9$, 2.1, 3.0, all at amp $= 10^{-4}$.

10.2 What it takes to clump

The ladder teaches three lessons (Fig. 8). First, $4\pi G > 2$ alone does not clump: the winding wave only *transits* its unstable window and bounces, and its quadratic self-coupling deposits only a whisper on the non-winding axisymmetric $(k_x, k_y) = (\pm 1, 0)$ mode (the $k_y = 0$ beat carries $k_x = 2k_x(t)$, which sweeps past the discrete mode instead of parking on it) — the $4\pi G = 3$ runs stay bounded even for 16 time units. Second, collapse requires the *crest itself* to become Jeans-unstable during the window: the local $\lambda_J = 2\pi c_s/\sqrt{4\pi G\rho}$ must drop below the crest width. At $4\pi G = 5$, amp $= 0.05$ this misses by $\sim$15% (peak $2.70\rho_0$, re-expands); at $4\pi G = 6$, amp $= 0.2$ it tips over: fragmentation at $t \approx 4.5$, runaway to $\rho = 203\rho_0$, and a bound clump that persists to the end of the run. Third, resolution: the uniform-grid Truelove ceiling ($\lambda_J \geq 4\Delta x$) at $4\pi G = 6$ is $\rho \approx 171\rho_0$ — the collapsed clump *exceeds* it, so the uniform run is marginally under-resolved precisely where it matters.

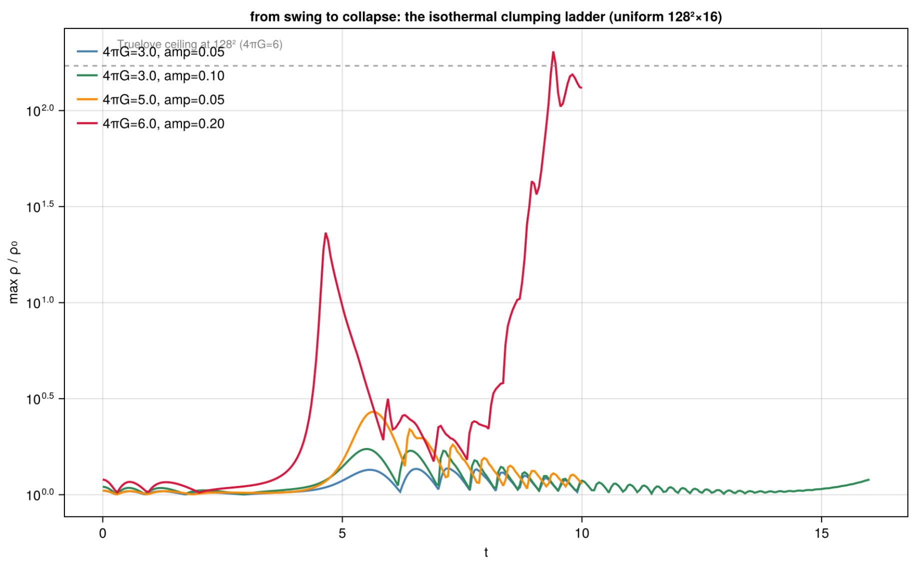

Figure 8: Maximum density vs. time for the nonlinear ladder: two bounces, a marginal miss, and the collapse, with the 128^2 Truelove ceiling marked.

10.3 Zooming on the clump under FARGO: rings, not patches

The clump is local, but with orbital advection on, the Phase-0 annular policy requires refinement in complete x_2 rings — so the AMR "zoom" refines the clump's x_1 columns at all y: a density criterion (`min_max` on `hydro_u_d`, `value_max=2`) flags the clump's blocks and the ring-sync completes the rings automatically. This is cheap here (one ring = 16 root blocks $\to$ 128 fine) and is what lets FARGO tolerate the clump's azimuthal drift. True local patches are legal only with `orbital_advection=false` at the measured $\sim 5.7\times$ timestep cost; per-ring bookkeeping for partially refined rings would be the Phase-3 alternative if deep zooms demand it. Input `tests/swing_selfgrav_clump_amr.athinput` runs the collapse with one refinement level chasing the crest and clump (level-1 Truelove ceiling $4\times$ higher, $\approx 684\rho_0$); the notebook (§15) compares the final density fields and mesh block structure of the uniform and AMR runs. The full run creates 1344 and destroys 1232 MeshBlocks — the rings form on the crest at $t \approx 4$, follow the fragmentation, and contract to a single ring tracking the clump (176 blocks at the end, 128 at level 1) — with zero multigrid warnings through the entire collapse. The density comparison carries the resolution lesson (Fig. 10): the uniform run peaks at $203\rho_0$, *above* its own Truelove ceiling ($171\rho_0$), while the AMR run peaks at $184\rho_0$ at $t = 9.9$ — comfortably inside its level-1 ceiling ($684\rho_0$) — so the uniform excess is under-resolution overshoot and only the AMR number is trustworthy. Both runs agree on formation time, location, and bulk structure of the *primary* clump. One open observation: the AMR run forms a *second* clump straddling the periodic y boundary (peak at $y = -3.105 \equiv +\pi$), absent in the uniform run — at the same x and separated by $\Delta y = \pi$, the wavelength of the k_y=2 second harmonic, which at $4\pi G = 6$ is itself Jeans-unstable ($\omega^2 = -1$) and quadratically seeded by the primary. This points to real fragmentation that the refined run resolves and the uniform grid merges, not a boundary artifact (the y wrap is plain periodic and translation-invariant). The 256^2 uniform discriminator run **settles it**: at matched fine resolution the same two-site structure appears — and an **FFT-solver twin** (single-rank; the FFT solver's scope) reproduces it *across solvers*: linear-phase $d_{\cos}$ agreement at 5×10^{-8}, and at t=10 the same two fragments within a third of a cell in position (boundary knot: FFT $41.9\rho_0$ at $(-0.01, -2.86)$ vs. multigrid $40.4\rho_0$ at $(-0.01, -2.88)$; companion 23.7 vs. 23.4) after five nonlinear dynamical times. Extended to t=14, the FFT twin also goes winner-takes-all at the same boundary site with the companion's site evacuated, and its runaway *clock* matches multigrid to 0.05 time units (the ρ=100 crossings); the two runs then part ways only *beyond* the Truelove ceiling — FFT saturates near it ($\sim 600\rho_0$, a grid-scale bounce) while multigrid shoots to $2.6 \times 10^5\rho_0$ — a clean, solver-based demonstration that past-ceiling densities are resolution artifacts in either code (Fig. 11) — — the dominant knot straddling $y = \pm\pi$ at $x \approx 0$ plus a center-side companion — confirming the k_y=2-harmonic fragmentation as physical and the 128^2 single clump as the under-resolved outlier. The properly resolved runs also collapse *later* (peak $40\rho_0$ at t=10, still contracting, vs. the 128^2 run's overshot 131): under-resolution both merges fragments and accelerates collapse. Extending the 256^2 run to t=14 at 64 ranks (the run that exposed and then validated the shear-exchange consistency fix) supplies the endpoint: the dominant knot undergoes full isothermal runaway to $2.6 \times 10^5\rho_0$ — 375× past its Truelove ceiling, deep into the unresolved regime — while consuming the center companion entirely (its site is a $0.4\rho_0$ void at the end). Absent sink particles or a floor, that runaway is the physical endpoint of isothermal collapse and the natural stopping criterion for uniform-grid studies; the multigrid solver held machine-clean through the entire quarter-million density contrast, ending exactly on a shear-periodic instant with zero warnings. The production lesson stands: chase fragmentation with AMR, and stop (or sink) at the Truelove ceiling. Figure 9 shows the three runs' winding-wave amplitudes $d_{\cos}/A$ — they share the linear swing and part ways at fragmentation ($t \approx 4.5$), where collapse hands energy to other harmonics — and Figure 10 their final states with the MeshBlock structure.

10.4 Why $d_{\cos} = \delta$: the projection diagnostic

The history diagnostic used throughout the swing program,

$$d_{\cos}(t) = \frac{2}{V}\int\left(\frac{\rho}{\rho_0} - 1\right)\cos\bigl(\mathbf{k}(t)\cdot\mathbf{x}\bigr)\,dV, \qquad \mathbf{k}(t) = \bigl(k_{x0} + q\Omega k_y t,\ k_y,\ 0\bigr), \tag{11}$$

equals the wave's density amplitude $\delta(t)$ exactly in the linear regime. Three steps:

(i) Linear disturbances wind. Linearizing about $\mathbf{v}_0 = -q\Omega x\,\hat{y}$ and passing to sheared coordinates $y' = y + q\Omega t\,x$ removes the background's explicit x-dependence, so plane waves in (x, y') remain plane waves for all time; mapped back, $e^{i(k_{x0}x + k_y y')} = e^{i\,\mathbf{k}(t)\cdot\mathbf{x}}$ with the winding $k_x(t)$ above. A single shwave is therefore exactly $\rho/\rho_0 = 1 + \delta(t)\cos\theta + B(t)\sin\theta$, $\theta \equiv \mathbf{k}(t)\cdot\mathbf{x}$.

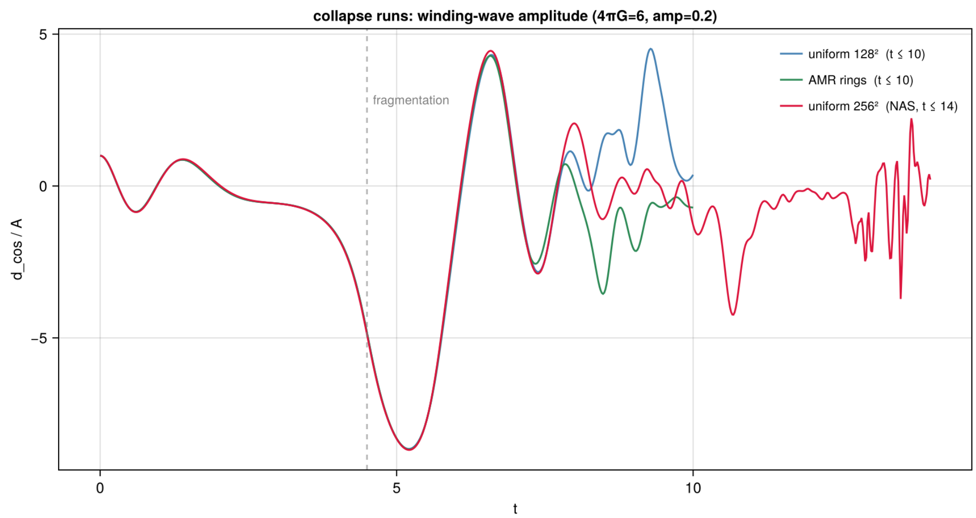

Figure 9: Winding-wave amplitude $d_{\cos}/A$ for the three collapse runs. The projection tracks only the $k(t)$ wave component, so the departure from the common curve is a clock for when fragmentation takes over.

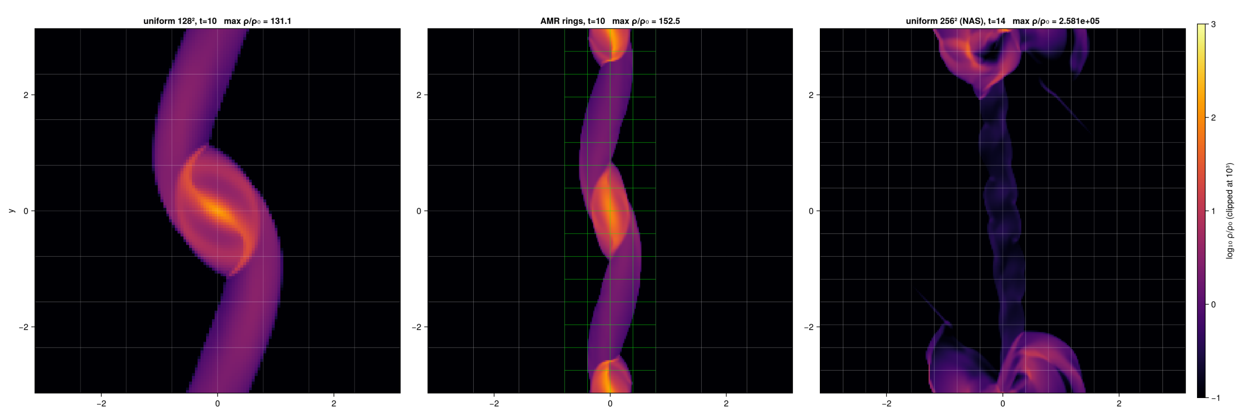

Figure 10: Final states of the three collapse runs ($\log_{10}\rho$, clipped at 10^3; MeshBlock outlines drawn, green = level 1): uniform 128^2 and the AMR run at $t = 10$, and the 256^2 NAS run at $t = 14$, deep in the winner-takes-all runaway with the companion's site evacuated.

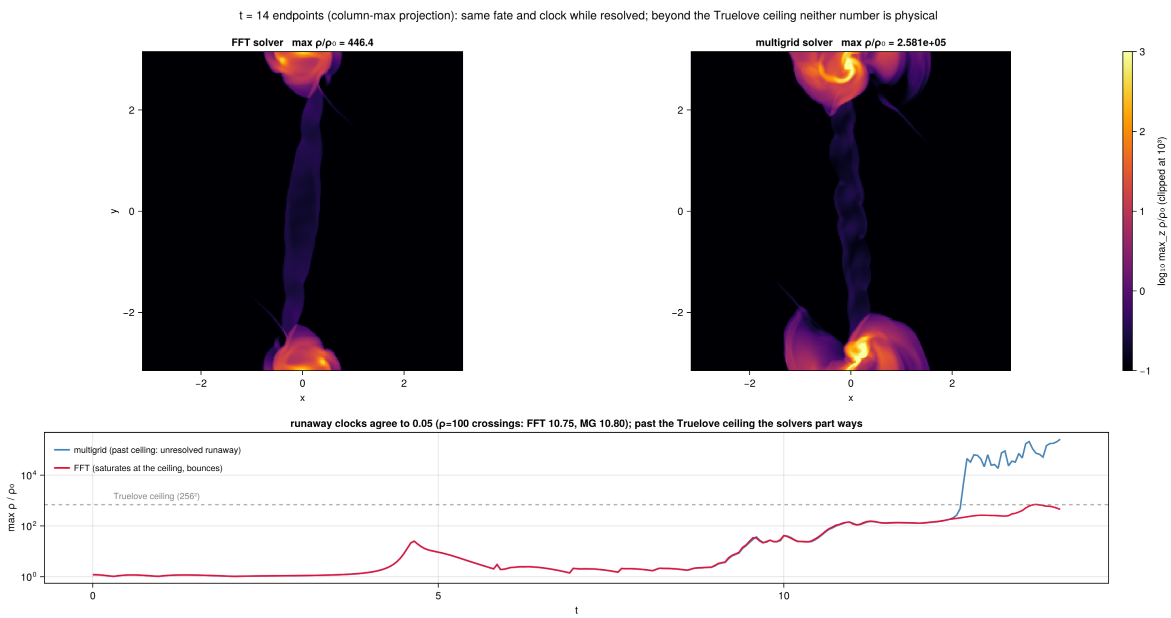

Figure 11: $t = 14$ endpoints of the 256^2 collapse under the two solvers (density panels are column-maximum projections over z — a mid-plane slice hides the multigrid core, which sits off-plane at $z = -0.21$): same fate (boundary knot wins, companion consumed) and runaway clocks agreeing to 0.05 while resolved; beyond the Truelove ceiling (dashed) the solvers diverge qualitatively — neither number is physical there.

(ii) The cosine channel is closed, and $B \equiv 0$. Differentiating $\delta \cos\theta$ at fixed $\mathbf{x}$ produces an awkward term $-\delta\,(q\Omega k_y x)\sin\theta$ from the winding of k_x; the background advection $\mathbf{v}_0 \cdot \nabla\rho_1 = +\,q\Omega k_y x\,\delta\sin\theta$ cancels it *exactly* — the winding wavevector is precisely the choice that lets the background shear advect the phase. What survives is the algebraic system quoted in the pgen header,

$$\dot{\delta} = -(k_x w_x + k_y w_y), \quad \dot{w}_x = 2\Omega w_y + k_x C\delta, \quad \dot{w}_y = -(2-q)\Omega w_x + k_y C\delta, \quad C = c_s^2 + \frac{4\pi G\rho_0}{D}, \tag{12}$$

for the phase assignment $\rho_1 \propto \cos\theta$, $(v_x, v_y) \propto \sin\theta$ (D the discrete Poisson eigenvalue, $\to -k^2$ in the continuum). This set is *closed*: cosine density couples only to sine velocities and back. The initial data ($\rho_1 = A\cos\theta$, $\mathbf{v}_1 = 0$) lies inside it, so the orthogonal channel is never excited and $B(t) = 0$ identically — which is why $\max|d_{\sin}|/A \sim 10^{-13}$ is such a sharp phase diagnostic: theory keeps that channel exactly empty.

(iii) The projection is exact, including on the grid. Inserting the ansatz, $d_{\cos} = (2\delta/V)\int\cos^2\theta\,dV = (\delta/V)\int(1+\cos 2\theta)\,dV$. The $\cos 2\theta$ term dies by the y-integral alone: 2θ contains $2k_y y$ with k_y an integer box mode, so $\int_0^{L_y}\cos(\cdot + 2k_y y)\,dy = 0$ *regardless of the instantaneous, generally non-resonant value of $k_x(t)$* — the subtle point; y-periodicity suffices. The same kills $\int\cos\theta\sin\theta\,dV$, making $d_{\cos}$ and $d_{\sin}$ independent. Hence $d_{\cos} = \delta(t)$, and normalizing by A gives the amplification factor. On a uniform grid the y-sum is $\sum_j e^{4\pi i n_y j/N_y} = 0$ exactly whenever $2n_y \not\equiv 0 \pmod{N_y}$, so the projection is exact on the grid too — the phase lock can reach machine precision rather than truncation level. Once a run goes nonlinear the field is no longer a single shwave and $d_{\cos}$ reverts to the amplitude of the $\mathbf{k}(t)$ Fourier component alone (the departure at fragmentation in Fig. 9).